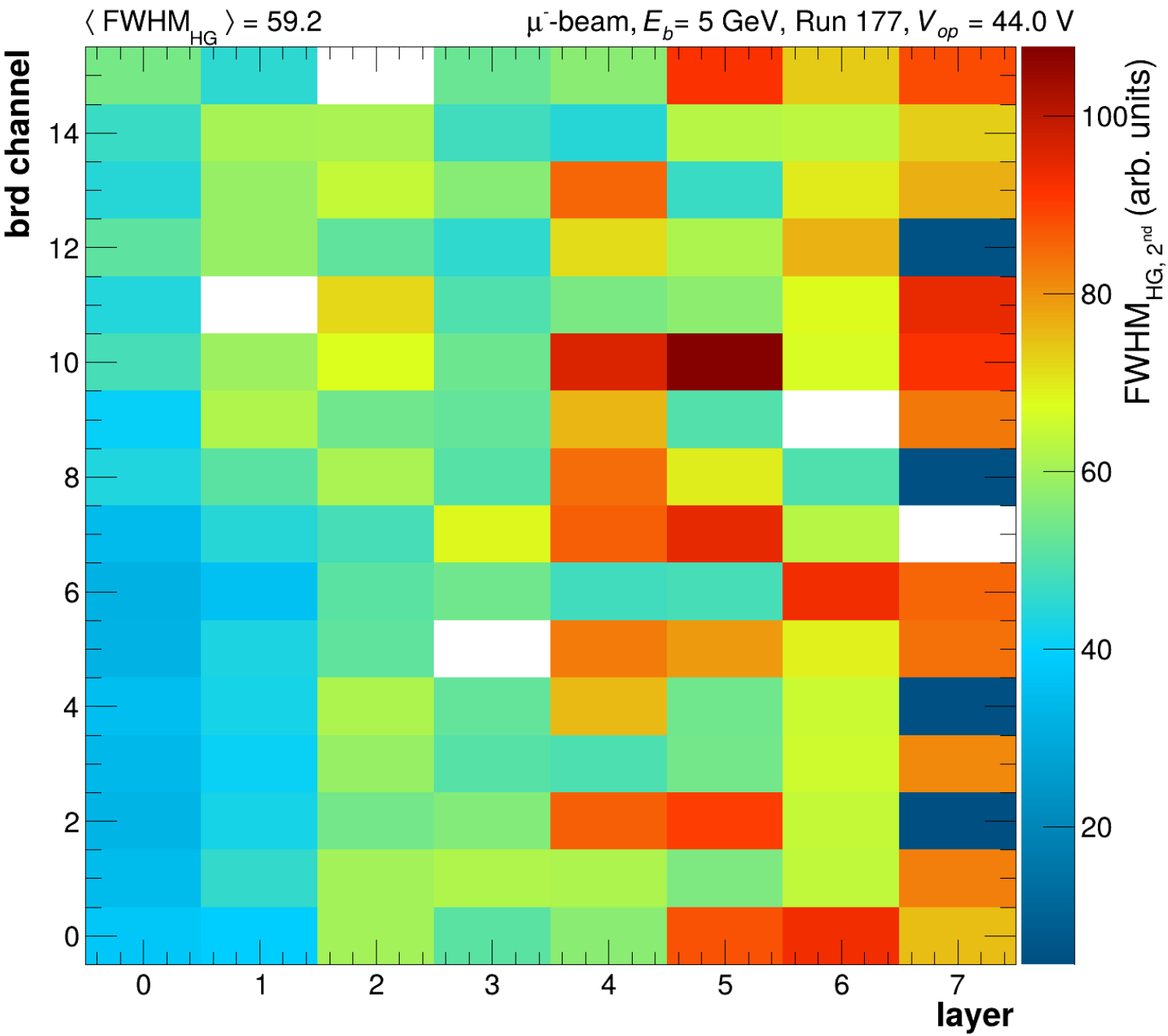
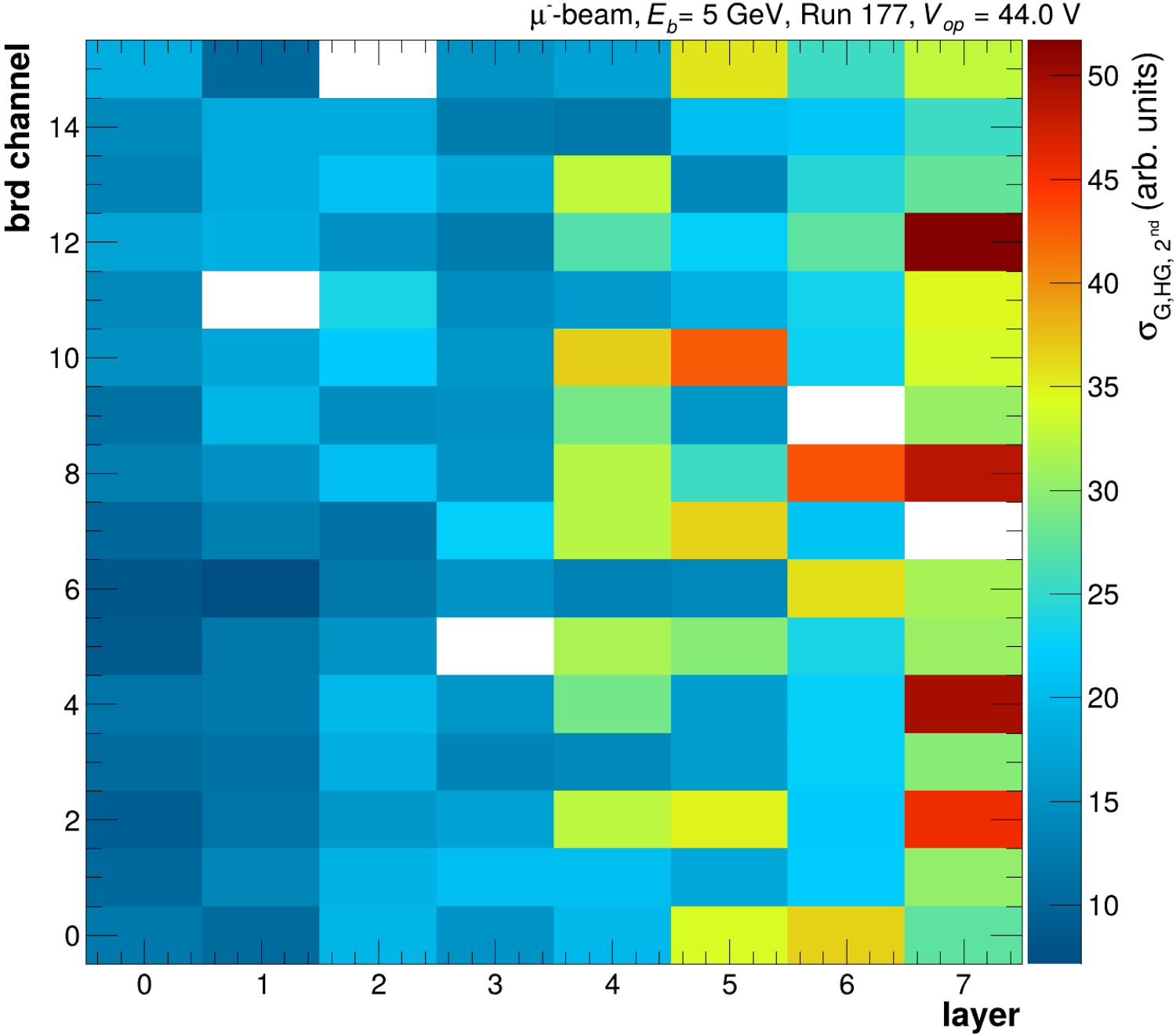
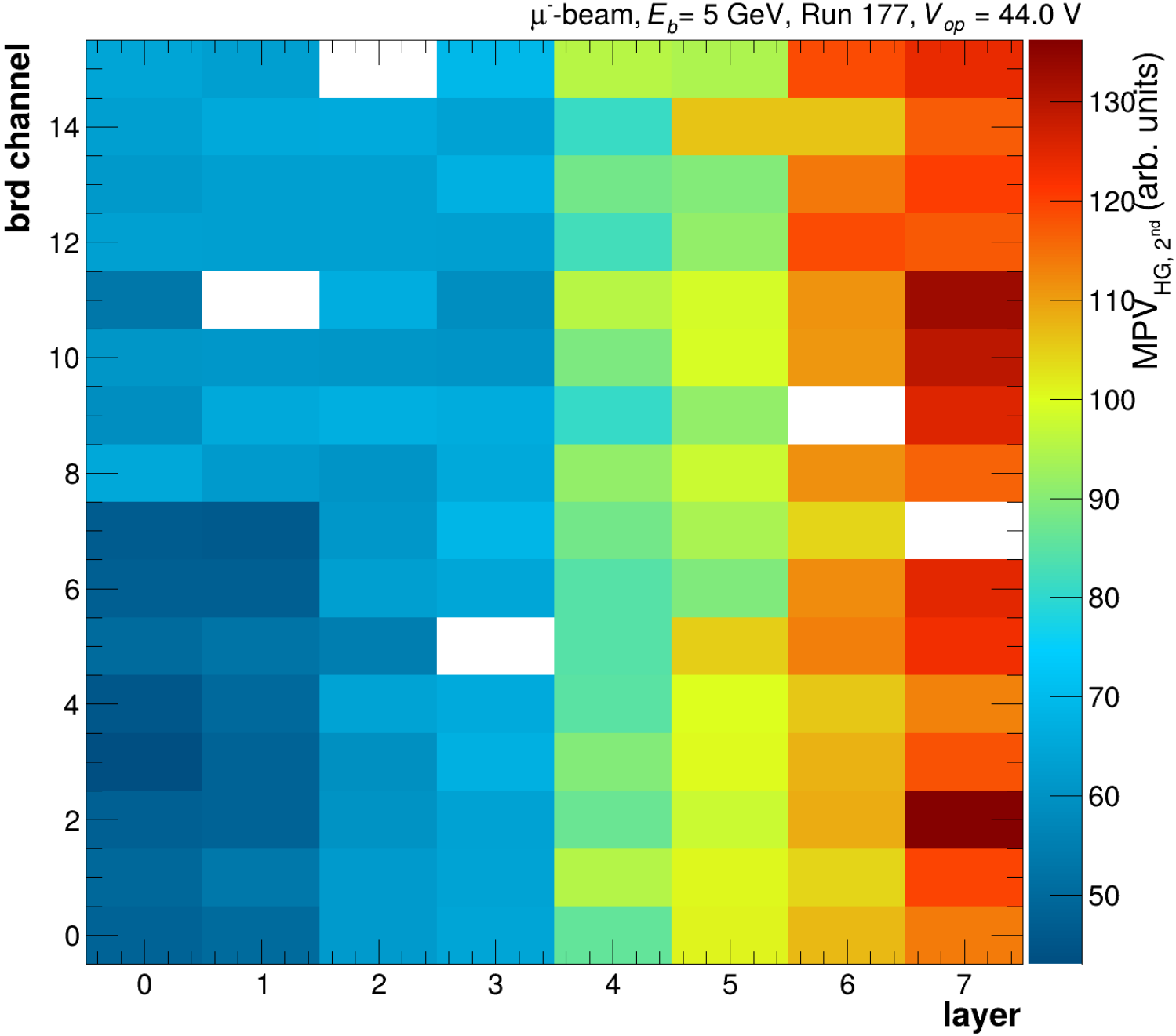
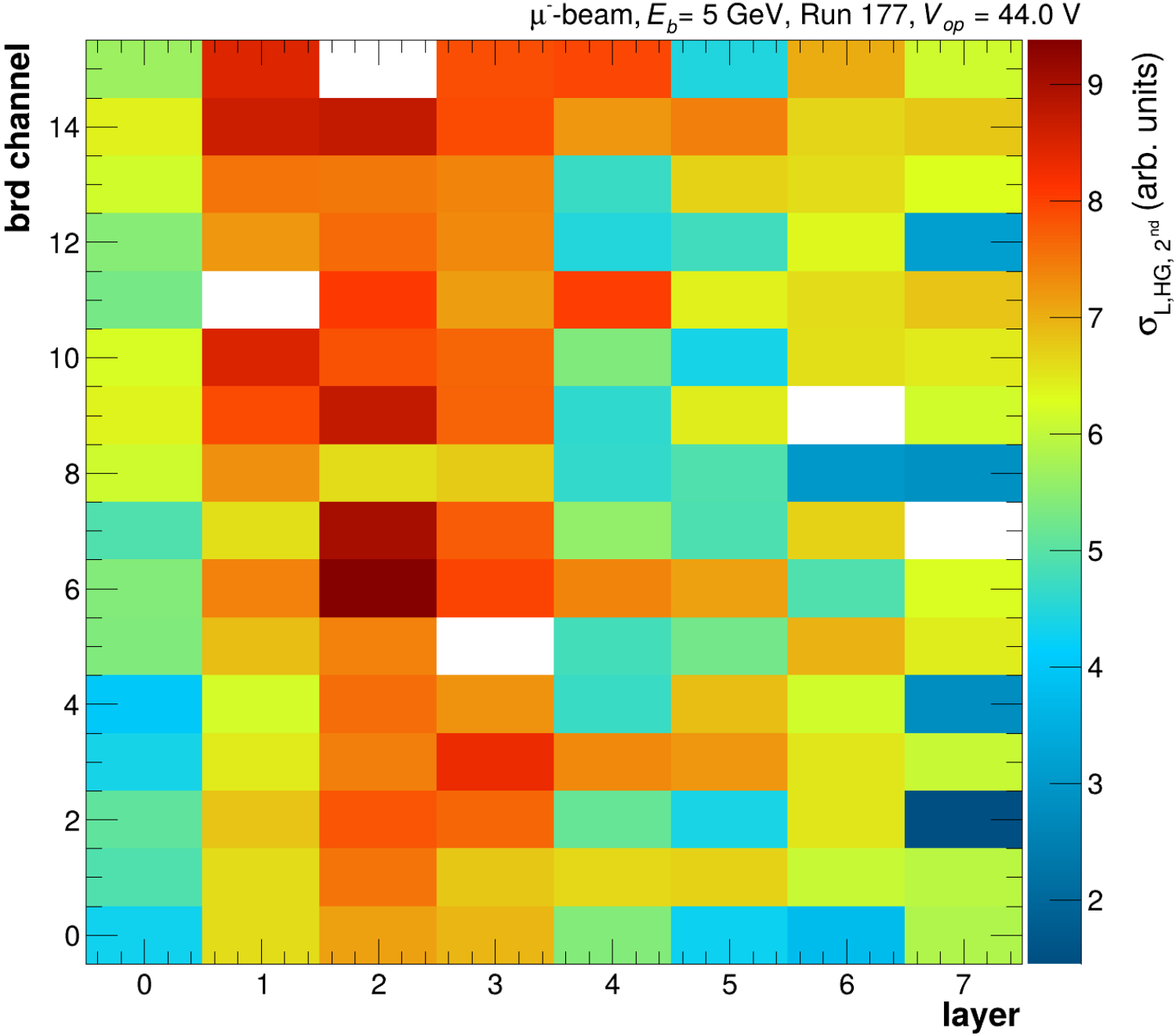
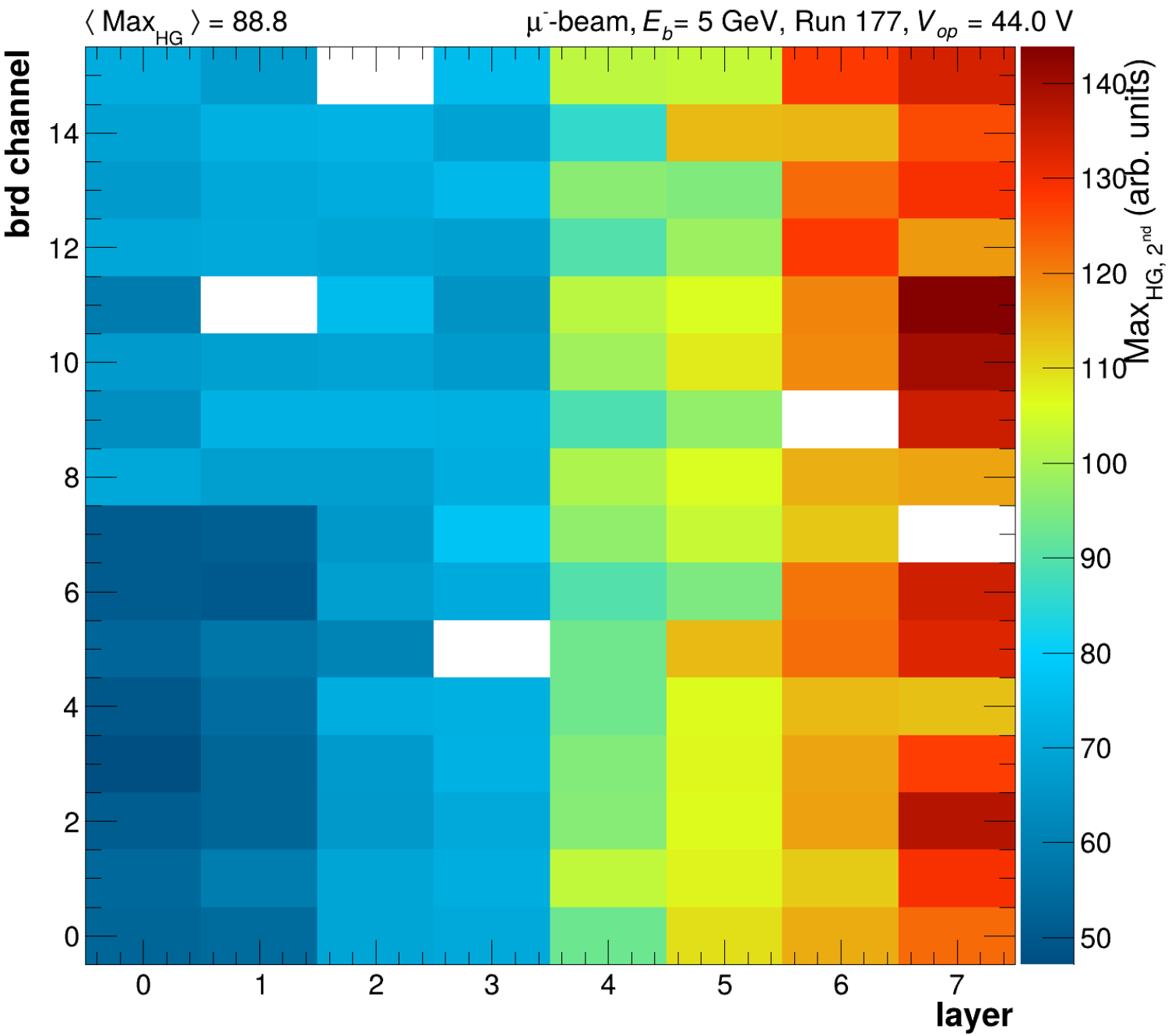
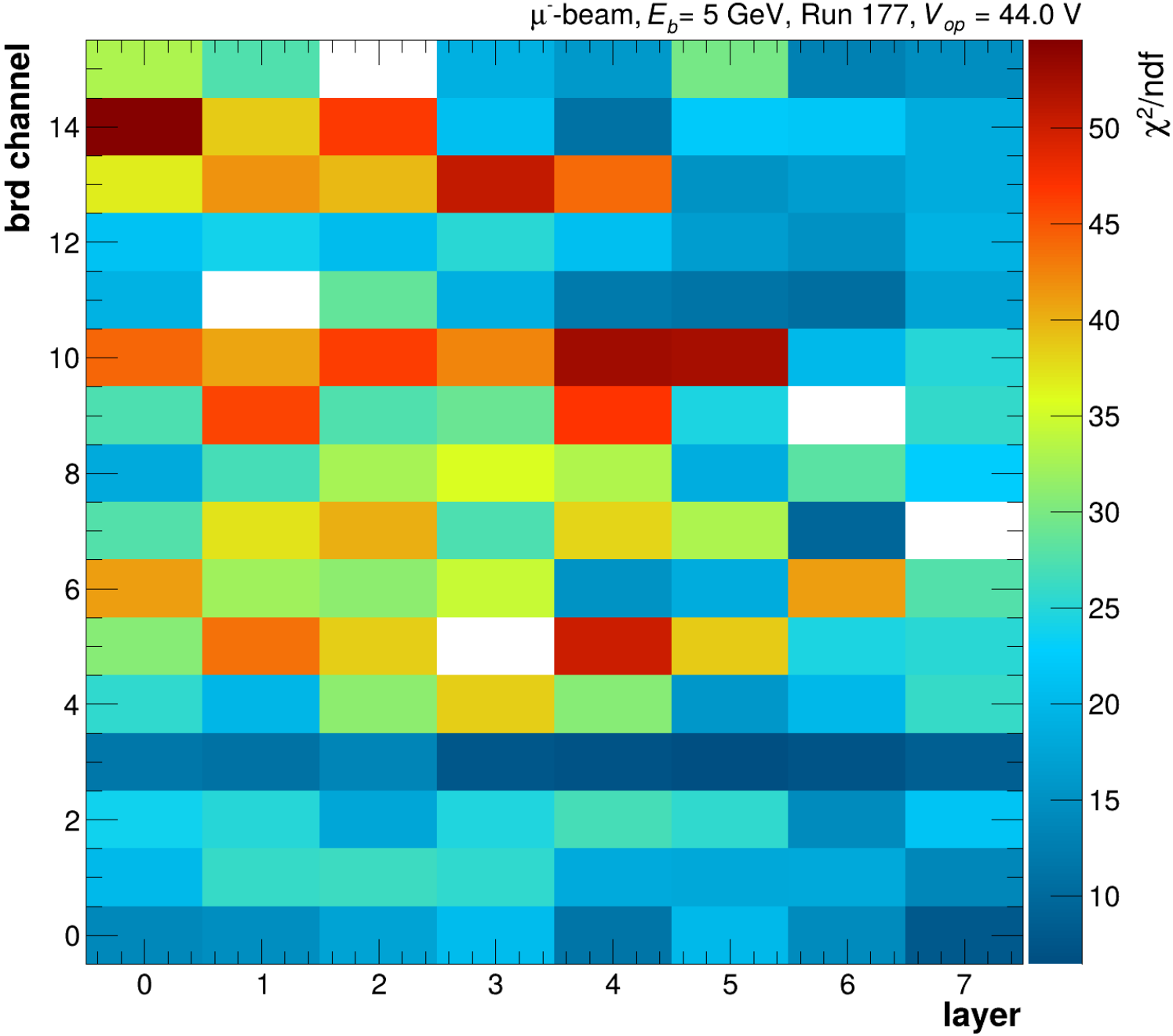

⟨ FWHM$_{HG}$ ⟩ = 59.2
μ$^-$-beam, E_b= 5 GeV, Run 177, V_{op} = 44.0 V
brd channel
layer
FWHM$_{HG,\ 2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
20
40
60
80
100

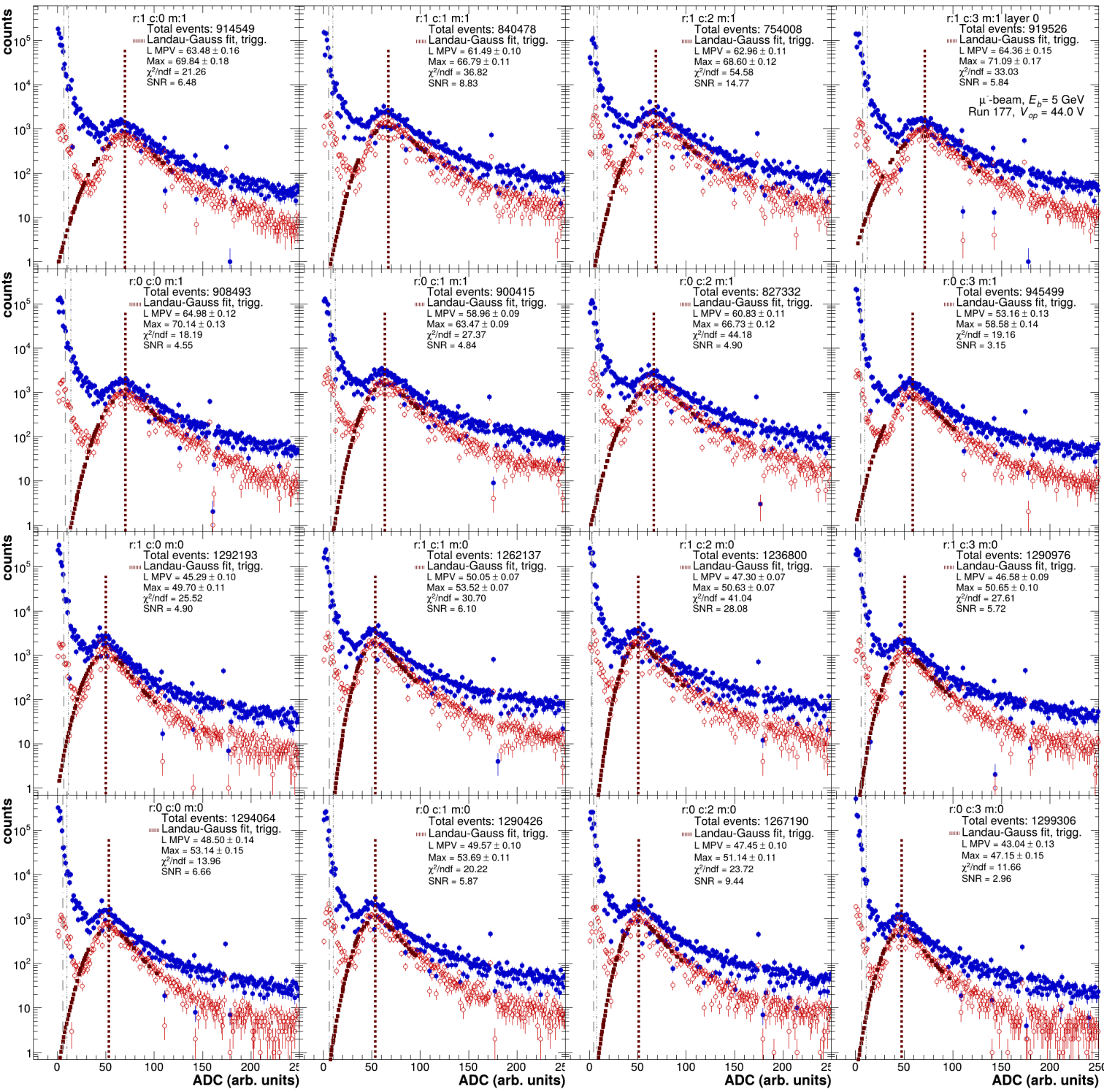

μ⁻-beam, E_b= 5 GeV, Run 177, V_{op} = 44.0 V
brd channel
layer
$\sigma_{G,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
10
15
20
25
30
35
40
45
50

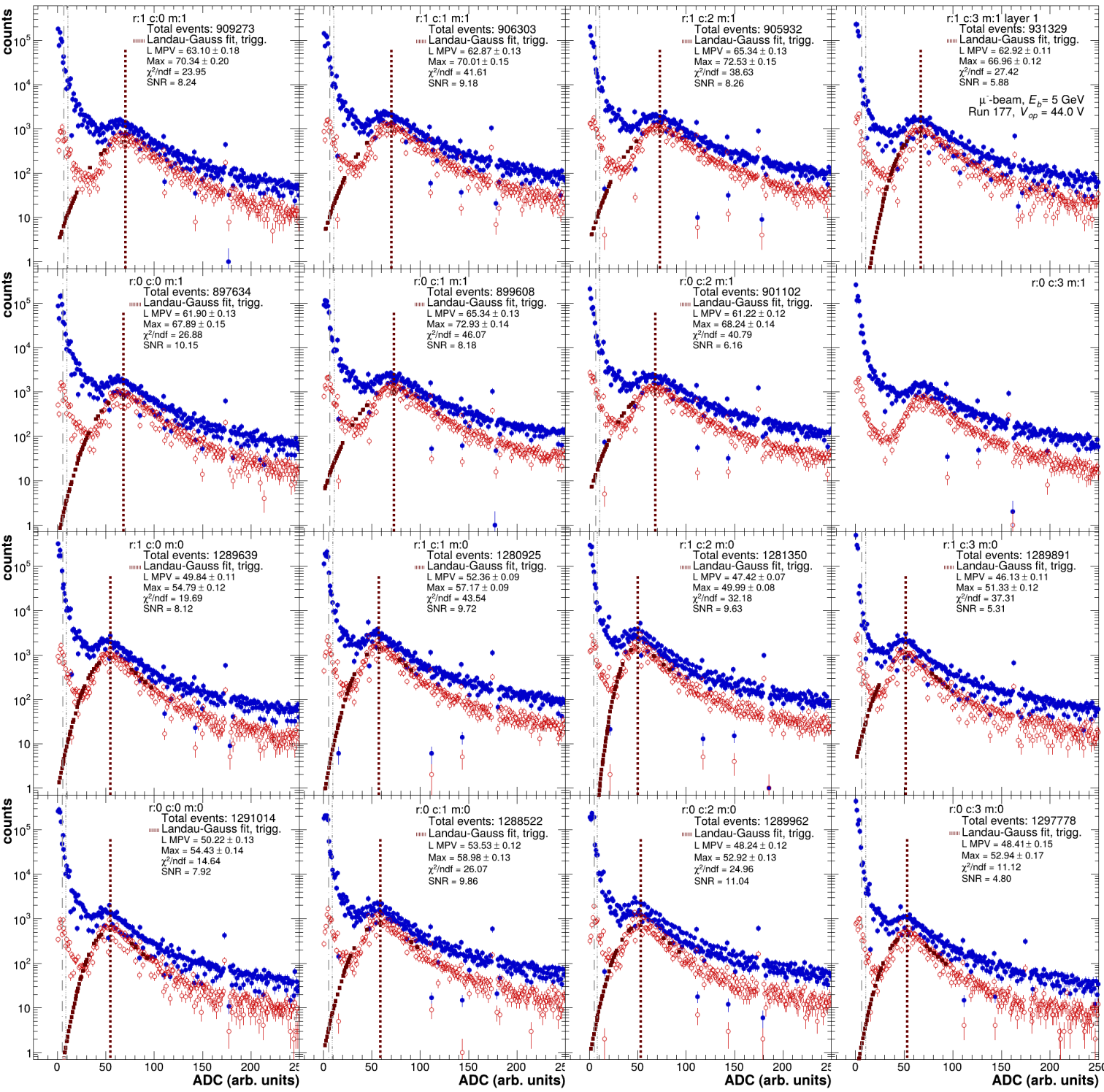

μ⁻-beam, E_b= 5 GeV, Run 177, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
MPV$_{HG,\,2^{nd}}$ (arb. units)
50
60
70
80
90
100
110
120
130

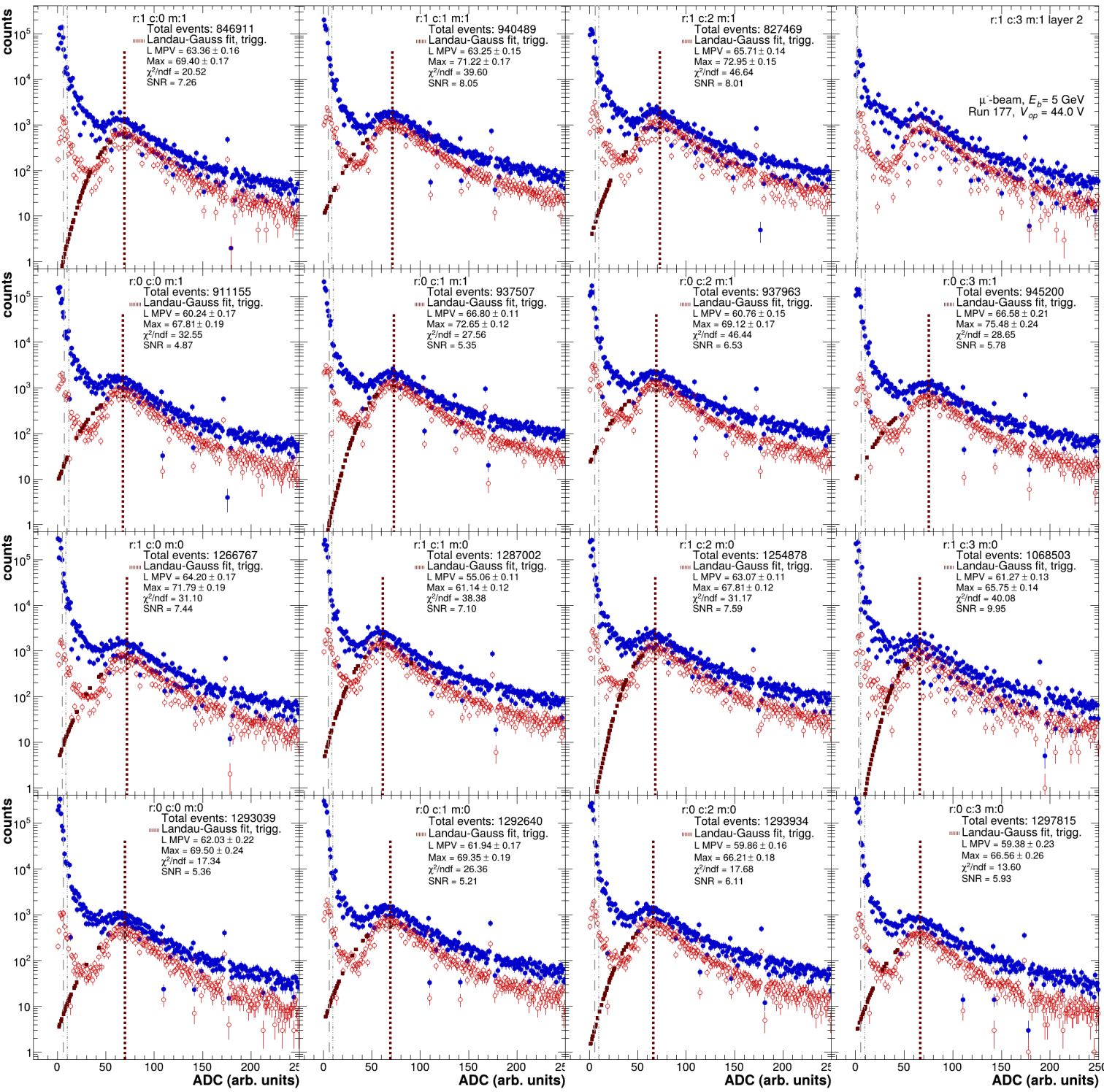

μ⁻-beam, E_b= 5 GeV, Run 177, V_{op} = 44.0 V
brd channel
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
1
3
5
7
9

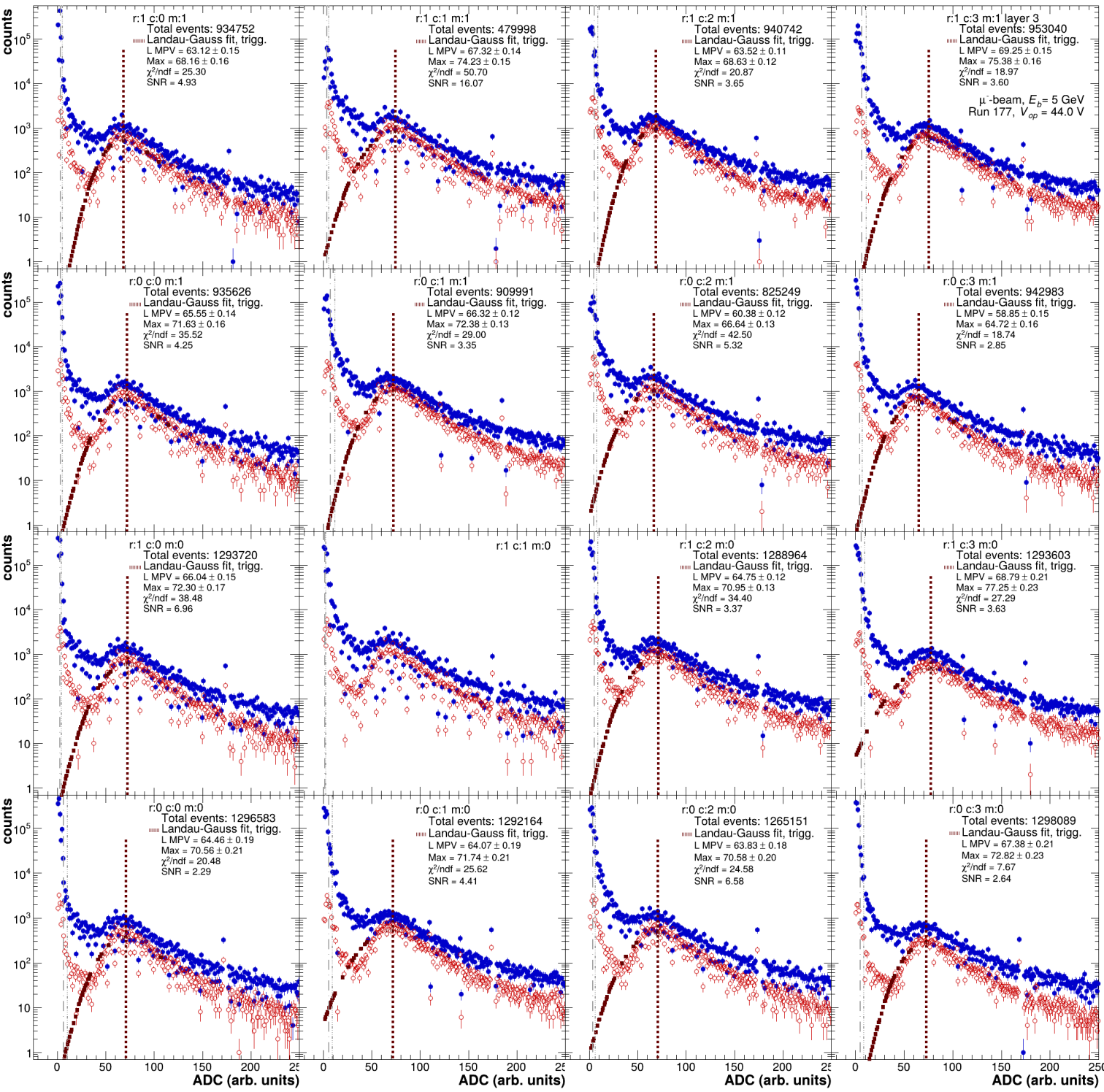

⟨ Max_HG ⟩ = 88.8
μ⁻-beam, E_b = 5 GeV, Run 177, V_op = 44.0 V
brd channel
layer
Max_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
50
60
70
80
90
100
110
120
130
140

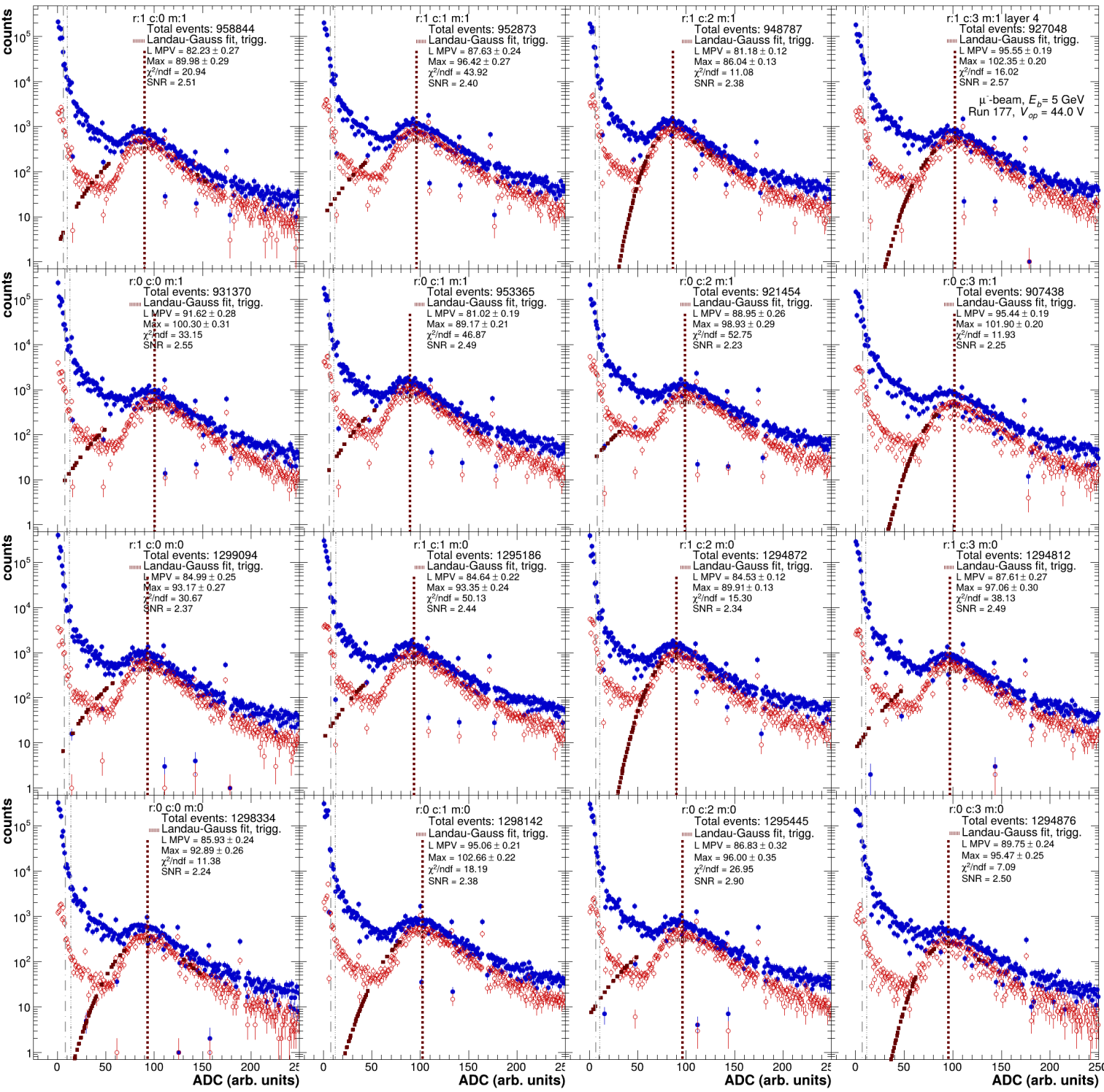

μ⁻-beam, E_b= 5 GeV, Run 177, V_{op} = 44.0 V
brd channel
layer
χ^2/ndf
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
10
15
20
25
30
35
40
45
50

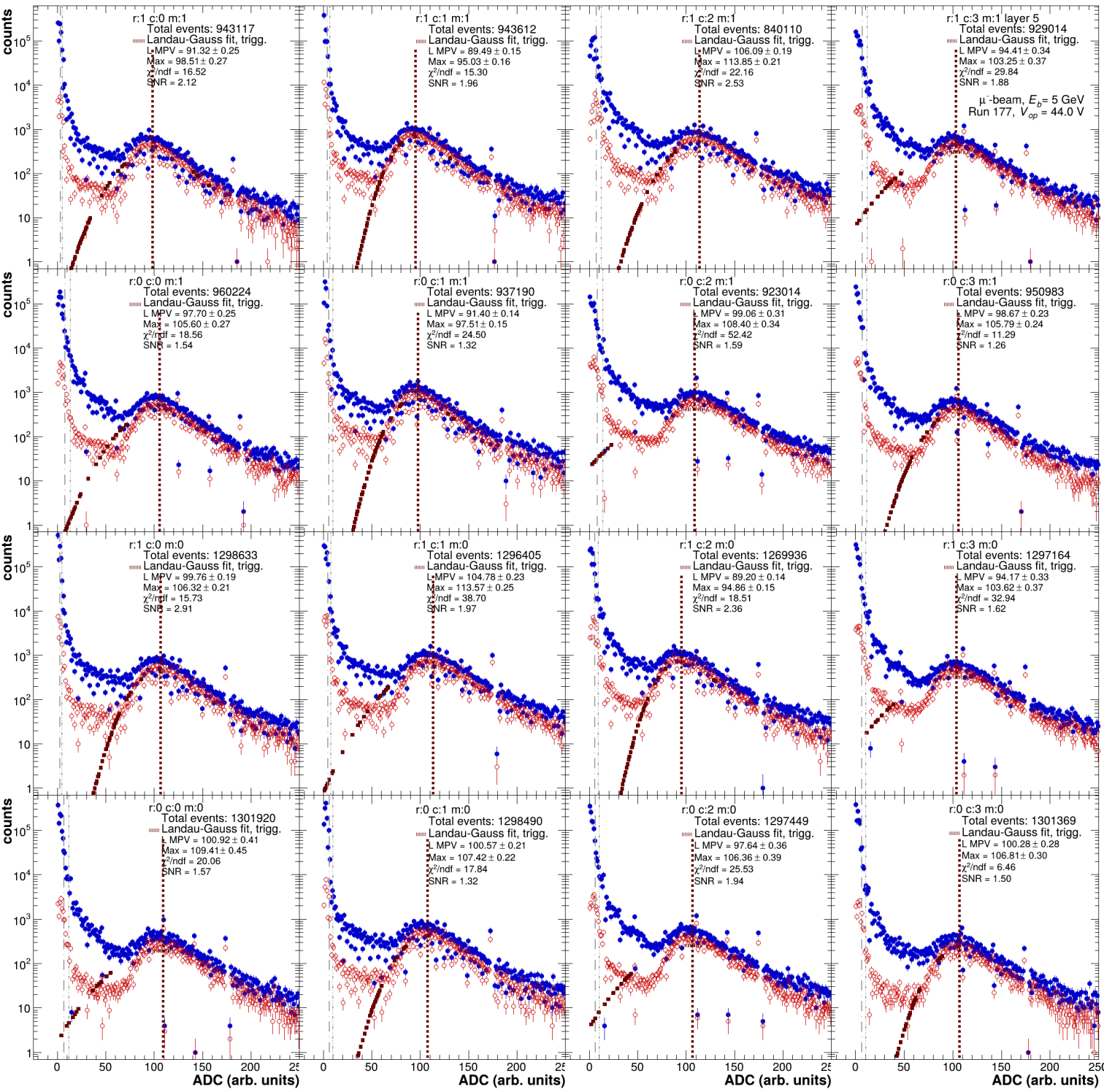

r:1 c:0 m:1
Total events: 914549
Landau-Gauss fit, trigg.
L MPV = 63.48 ± 0.16
Max = 69.84 ± 0.18
χ²/ndf = 21.26
SNR = 6.48
r:1 c:1 m:1
Total events: 840478
Landau-Gauss fit, trigg.
L MPV = 61.49 ± 0.10
Max = 66.79 ± 0.11
χ²/ndf = 36.82
SNR = 8.83
r:1 c:2 m:1
Total events: 754008
Landau-Gauss fit, trigg.
L MPV = 62.96 ± 0.11
Max = 68.60 ± 0.12
χ²/ndf = 54.58
SNR = 14.77
r:1 c:3 m:1 layer 0
Total events: 919526
Landau-Gauss fit, trigg.
L MPV = 64.36 ± 0.15
Max = 71.09 ± 0.17
χ²/ndf = 33.03
SNR = 5.84
μ⁻-beam, E_b = 5 GeV
Run 177, V_op = 44.0 V
r:0 c:0 m:1
Total events: 908493
Landau-Gauss fit, trigg.
L MPV = 64.98 ± 0.12
Max = 70.14 ± 0.13
χ²/ndf = 18.19
SNR = 4.55
r:0 c:1 m:1
Total events: 900415
Landau-Gauss fit, trigg.
L MPV = 58.96 ± 0.09
Max = 63.47 ± 0.09
χ²/ndf = 27.37
SNR = 4.84
r:0 c:2 m:1
Total events: 827332
Landau-Gauss fit, trigg.
L MPV = 60.83 ± 0.11
Max = 66.73 ± 0.12
χ²/ndf = 44.18
SNR = 4.90
r:0 c:3 m:1
Total events: 945499
Landau-Gauss fit, trigg.
L MPV = 53.16 ± 0.13
Max = 58.58 ± 0.14
χ²/ndf = 19.16
SNR = 3.15
r:1 c:0 m:0
Total events: 1292193
Landau-Gauss fit, trigg.
L MPV = 45.29 ± 0.10
Max = 49.70 ± 0.11
χ²/ndf = 25.52
SNR = 4.90
r:1 c:1 m:0
Total events: 1262137
Landau-Gauss fit, trigg.
L MPV = 50.05 ± 0.07
Max = 53.52 ± 0.07
χ²/ndf = 30.70
SNR = 6.10
r:1 c:2 m:0
Total events: 1236800
Landau-Gauss fit, trigg.
L MPV = 47.30 ± 0.07
Max = 50.63 ± 0.07
χ²/ndf = 41.04
SNR = 28.08
r:1 c:3 m:0
Total events: 1290976
Landau-Gauss fit, trigg.
L MPV = 46.58 ± 0.09
Max = 50.65 ± 0.10
χ²/ndf = 27.61
SNR = 5.72
r:0 c:0 m:0
Total events: 1294064
Landau-Gauss fit, trigg.
L MPV = 48.50 ± 0.14
Max = 53.14 ± 0.15
χ²/ndf = 13.96
SNR = 6.66
r:0 c:1 m:0
Total events: 1290426
Landau-Gauss fit, trigg.
L MPV = 49.57 ± 0.10
Max = 53.69 ± 0.11
χ²/ndf = 20.22
SNR = 5.87
r:0 c:2 m:0
Total events: 1267190
Landau-Gauss fit, trigg.
L MPV = 47.45 ± 0.10
Max = 51.14 ± 0.11
χ²/ndf = 23.72
SNR = 9.44
r:0 c:3 m:0
Total events: 1299306
Landau-Gauss fit, trigg.
L MPV = 43.04 ± 0.13
Max = 47.15 ± 0.15
χ²/ndf = 11.66
SNR = 2.96
counts
ADC (arb. units)

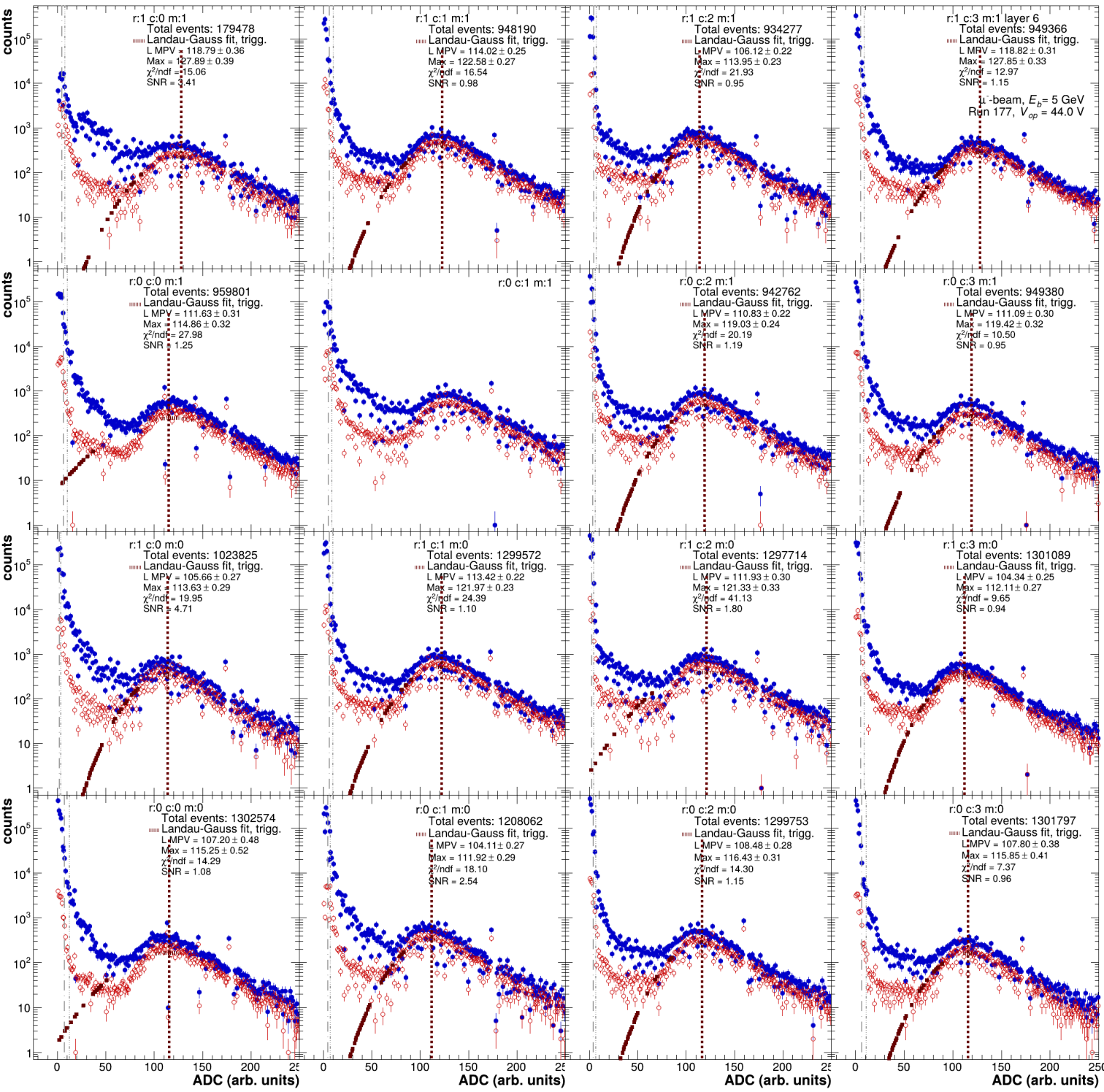

r:1 c:0 m:1
Total events: 909273
Landau-Gauss fit, trigg.
L MPV = 63.10 ± 0.18
Max = 70.34 ± 0.20
χ²/ndf = 23.95
SNR = 8.24
r:1 c:1 m:1
Total events: 906303
Landau-Gauss fit, trigg.
L MPV = 62.87 ± 0.13
Max = 70.01 ± 0.15
χ²/ndf = 41.61
SNR = 9.18
r:1 c:2 m:1
Total events: 905932
Landau-Gauss fit, trigg.
L MPV = 65.34 ± 0.13
Max = 72.53 ± 0.15
χ²/ndf = 38.63
SNR = 8.26
r:1 c:3 m:1 layer 1
Total events: 931329
Landau-Gauss fit, trigg.
L MPV = 62.92 ± 0.11
Max = 66.96 ± 0.12
χ²/ndf = 27.42
SNR = 5.88
μ⁻-beam, E_b = 5 GeV
Run 177, V_op = 44.0 V
r:0 c:0 m:1
Total events: 897634
Landau-Gauss fit, trigg.
L MPV = 61.90 ± 0.13
Max = 67.89 ± 0.15
χ²/ndf = 26.88
SNR = 10.15
r:0 c:1 m:1
Total events: 899608
Landau-Gauss fit, trigg.
L MPV = 65.34 ± 0.13
Max = 72.93 ± 0.14
χ²/ndf = 46.07
SNR = 8.18
r:0 c:2 m:1
Total events: 901102
Landau-Gauss fit, trigg.
L MPV = 61.22 ± 0.12
Max = 68.24 ± 0.14
χ²/ndf = 40.79
SNR = 6.16
r:0 c:3 m:1
r:1 c:0 m:0
Total events: 1289639
Landau-Gauss fit, trigg.
L MPV = 49.84 ± 0.11
Max = 54.79 ± 0.12
χ²/ndf = 19.69
SNR = 8.12
r:1 c:1 m:0
Total events: 1280925
Landau-Gauss fit, trigg.
L MPV = 52.36 ± 0.09
Max = 57.17 ± 0.09
χ²/ndf = 43.54
SNR = 9.72
r:1 c:2 m:0
Total events: 1281350
Landau-Gauss fit, trigg.
L MPV = 47.42 ± 0.07
Max = 49.99 ± 0.08
χ²/ndf = 32.18
SNR = 9.63
r:1 c:3 m:0
Total events: 1289891
Landau-Gauss fit, trigg.
L MPV = 46.13 ± 0.11
Max = 51.33 ± 0.12
χ²/ndf = 37.31
SNR = 5.31
r:0 c:0 m:0
Total events: 1291014
Landau-Gauss fit, trigg.
L MPV = 50.22 ± 0.13
Max = 54.43 ± 0.14
χ²/ndf = 14.64
SNR = 7.92
r:0 c:1 m:0
Total events: 1288522
Landau-Gauss fit, trigg.
L MPV = 53.53 ± 0.12
Max = 58.98 ± 0.13
χ²/ndf = 26.07
SNR = 9.86
r:0 c:2 m:0
Total events: 1289962
Landau-Gauss fit, trigg.
L MPV = 48.24 ± 0.12
Max = 52.92 ± 0.13
χ²/ndf = 24.96
SNR = 11.04
r:0 c:3 m:0
Total events: 1297778
Landau-Gauss fit, trigg.
L MPV = 48.41 ± 0.15
Max = 52.94 ± 0.17
χ²/ndf = 11.12
SNR = 4.80
counts
ADC (arb. units)

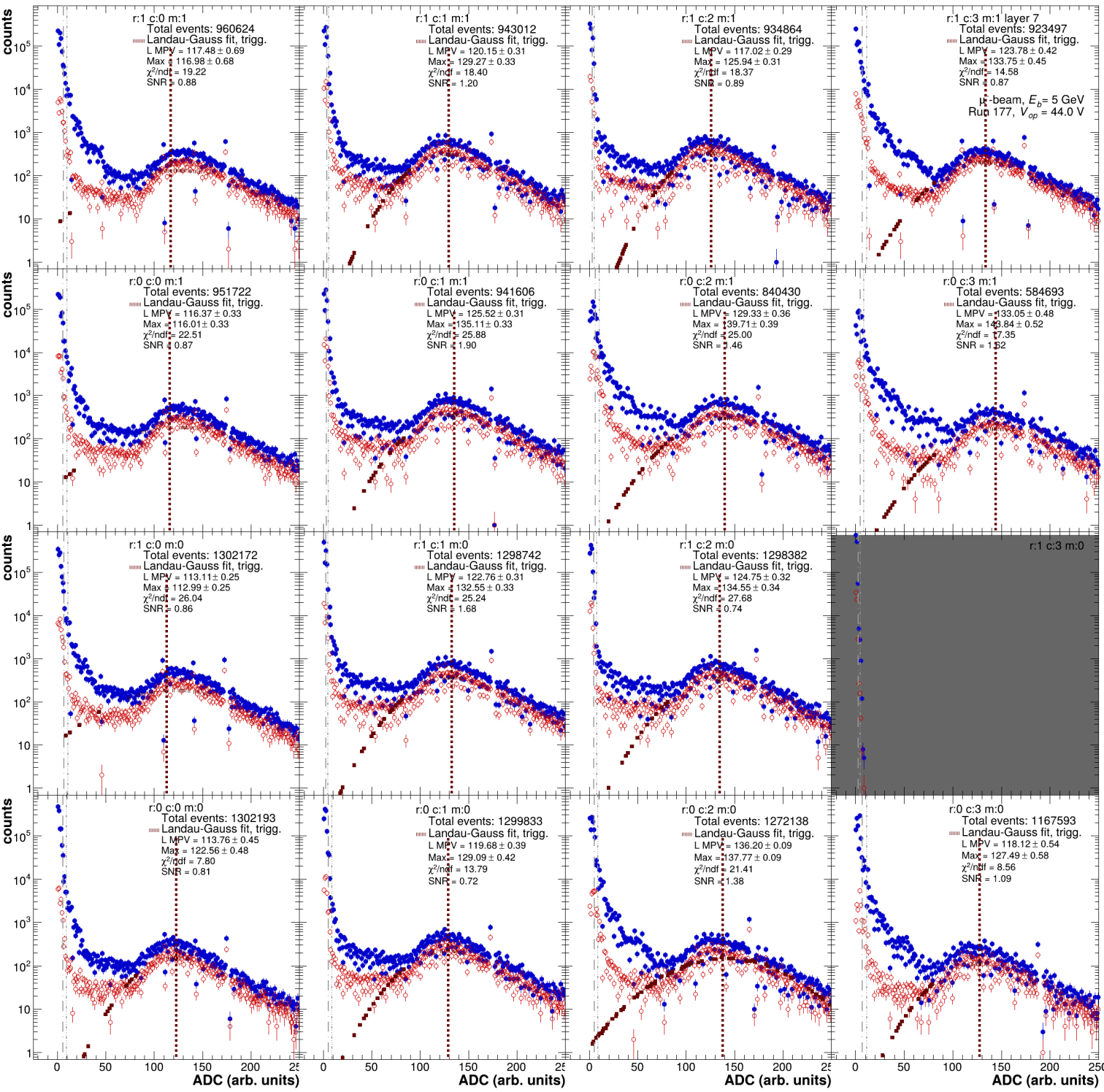

r:1 c:0 m:1
Total events: 846911
Landau-Gauss fit, trigg.
L MPV = 63.36 ± 0.16
Max = 69.40 ± 0.17
χ²/ndf = 20.52
SNR = 7.26
r:1 c:1 m:1
Total events: 940489
Landau-Gauss fit, trigg.
L MPV = 63.25 ± 0.15
Max = 71.22 ± 0.17
χ²/ndf = 39.60
SNR = 8.05
r:1 c:2 m:1
Total events: 827469
Landau-Gauss fit, trigg.
L MPV = 65.71 ± 0.14
Max = 72.95 ± 0.15
χ²/ndf = 46.64
SNR = 8.01
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b = 5 GeV
Run 177, V_op = 44.0 V
r:0 c:0 m:1
Total events: 911155
Landau-Gauss fit, trigg.
L MPV = 60.24 ± 0.17
Max = 67.81 ± 0.19
χ²/ndf = 32.55
SNR = 4.87
r:0 c:1 m:1
Total events: 937507
Landau-Gauss fit, trigg.
L MPV = 66.80 ± 0.11
Max = 72.65 ± 0.12
χ²/ndf = 27.56
SNR = 5.35
r:0 c:2 m:1
Total events: 937963
Landau-Gauss fit, trigg.
L MPV = 60.76 ± 0.15
Max = 69.12 ± 0.17
χ²/ndf = 46.44
SNR = 6.53
r:0 c:3 m:1
Total events: 945200
Landau-Gauss fit, trigg.
L MPV = 66.58 ± 0.21
Max = 75.48 ± 0.24
χ²/ndf = 28.65
SNR = 5.78
r:1 c:0 m:0
Total events: 1266767
Landau-Gauss fit, trigg.
L MPV = 64.20 ± 0.17
Max = 71.79 ± 0.19
χ²/ndf = 31.10
SNR = 7.44
r:1 c:1 m:0
Total events: 1287002
Landau-Gauss fit, trigg.
L MPV = 55.06 ± 0.11
Max = 61.14 ± 0.12
χ²/ndf = 38.38
SNR = 7.10
r:1 c:2 m:0
Total events: 1254878
Landau-Gauss fit, trigg.
L MPV = 63.07 ± 0.11
Max = 67.81 ± 0.12
χ²/ndf = 31.17
SNR = 7.59
r:1 c:3 m:0
Total events: 1068503
Landau-Gauss fit, trigg.
L MPV = 61.27 ± 0.13
Max = 65.75 ± 0.14
χ²/ndf = 40.08
SNR = 9.95
r:0 c:0 m:0
Total events: 1293039
Landau-Gauss fit, trigg.
L MPV = 62.03 ± 0.22
Max = 69.50 ± 0.24
χ²/ndf = 17.34
SNR = 5.36
r:0 c:1 m:0
Total events: 1292640
Landau-Gauss fit, trigg.
L MPV = 61.94 ± 0.17
Max = 69.35 ± 0.19
χ²/ndf = 26.36
SNR = 5.21
r:0 c:2 m:0
Total events: 1293934
Landau-Gauss fit, trigg.
L MPV = 59.86 ± 0.16
Max = 66.21 ± 0.18
χ²/ndf = 17.68
SNR = 6.11
r:0 c:3 m:0
Total events: 1297815
Landau-Gauss fit, trigg.
L MPV = 59.38 ± 0.23
Max = 66.56 ± 0.26
χ²/ndf = 13.60
SNR = 5.93
counts
ADC (arb. units)

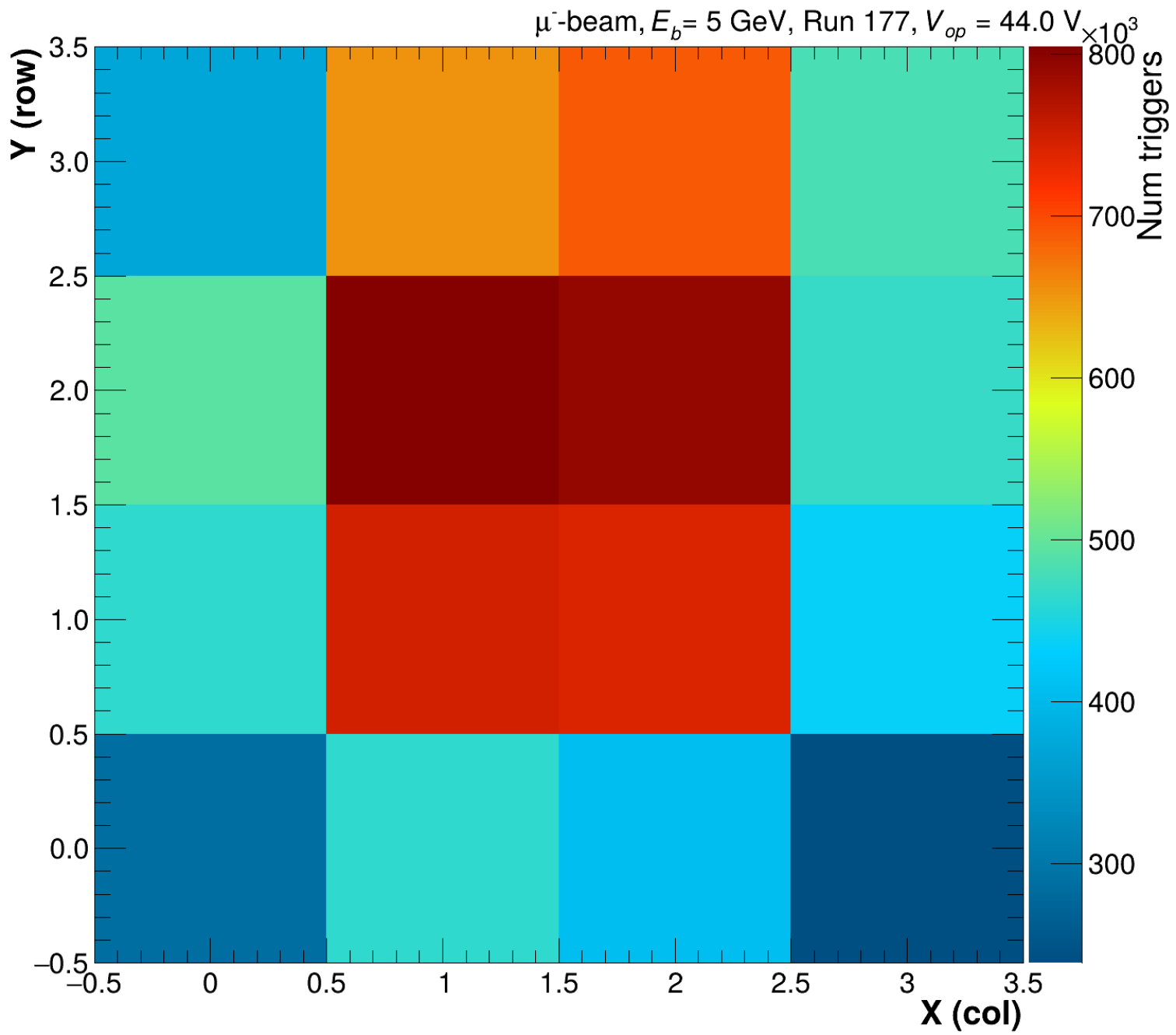

r:1 c:0 m:1
Total events: 934752
Landau-Gauss fit, trigg.
L MPV = 63.12 ± 0.15
Max = 68.16 ± 0.16
χ²/ndf = 25.30
SNR = 4.93
r:1 c:1 m:1
Total events: 479998
Landau-Gauss fit, trigg.
L MPV = 67.32 ± 0.14
Max = 74.23 ± 0.15
χ²/ndf = 50.70
SNR = 16.07
r:1 c:2 m:1
Total events: 940742
Landau-Gauss fit, trigg.
L MPV = 63.52 ± 0.11
Max = 68.63 ± 0.12
χ²/ndf = 20.87
SNR = 3.65
r:1 c:3 m:1 layer 3
Total events: 953040
Landau-Gauss fit, trigg.
L MPV = 69.25 ± 0.15
Max = 75.38 ± 0.16
χ²/ndf = 18.97
SNR = 3.60
μ⁻-beam, E_b = 5 GeV
Run 177, V_op = 44.0 V
r:0 c:0 m:1
Total events: 935626
Landau-Gauss fit, trigg.
L MPV = 65.55 ± 0.14
Max = 71.63 ± 0.16
χ²/ndf = 35.52
SNR = 4.25
r:0 c:1 m:1
Total events: 909991
Landau-Gauss fit, trigg.
L MPV = 66.32 ± 0.12
Max = 72.38 ± 0.13
χ²/ndf = 29.00
SNR = 3.35
r:0 c:2 m:1
Total events: 825249
Landau-Gauss fit, trigg.
L MPV = 60.38 ± 0.12
Max = 66.64 ± 0.13
χ²/ndf = 42.50
SNR = 5.32
r:0 c:3 m:1
Total events: 942983
Landau-Gauss fit, trigg.
L MPV = 58.85 ± 0.15
Max = 64.72 ± 0.16
χ²/ndf = 18.74
SNR = 2.85
r:1 c:0 m:0
Total events: 1293720
Landau-Gauss fit, trigg.
L MPV = 66.04 ± 0.15
Max = 72.30 ± 0.17
χ²/ndf = 38.48
SNR = 6.96
r:1 c:1 m:0
r:1 c:2 m:0
Total events: 1288964
Landau-Gauss fit, trigg.
L MPV = 64.75 ± 0.12
Max = 70.95 ± 0.13
χ²/ndf = 34.40
SNR = 3.37
r:1 c:3 m:0
Total events: 1293603
Landau-Gauss fit, trigg.
L MPV = 68.79 ± 0.21
Max = 77.25 ± 0.23
χ²/ndf = 27.29
SNR = 3.63
r:0 c:0 m:0
Total events: 1296583
Landau-Gauss fit, trigg.
L MPV = 64.46 ± 0.19
Max = 70.56 ± 0.21
χ²/ndf = 20.48
SNR = 2.29
r:0 c:1 m:0
Total events: 1292164
Landau-Gauss fit, trigg.
L MPV = 64.07 ± 0.19
Max = 71.74 ± 0.21
χ²/ndf = 25.62
SNR = 4.41
r:0 c:2 m:0
Total events: 1265151
Landau-Gauss fit, trigg.
L MPV = 63.83 ± 0.18
Max = 70.58 ± 0.20
χ²/ndf = 24.58
SNR = 6.58
r:0 c:3 m:0
Total events: 1298089
Landau-Gauss fit, trigg.
L MPV = 67.38 ± 0.21
Max = 72.82 ± 0.23
χ²/ndf = 7.67
SNR = 2.64
counts
ADC (arb. units)

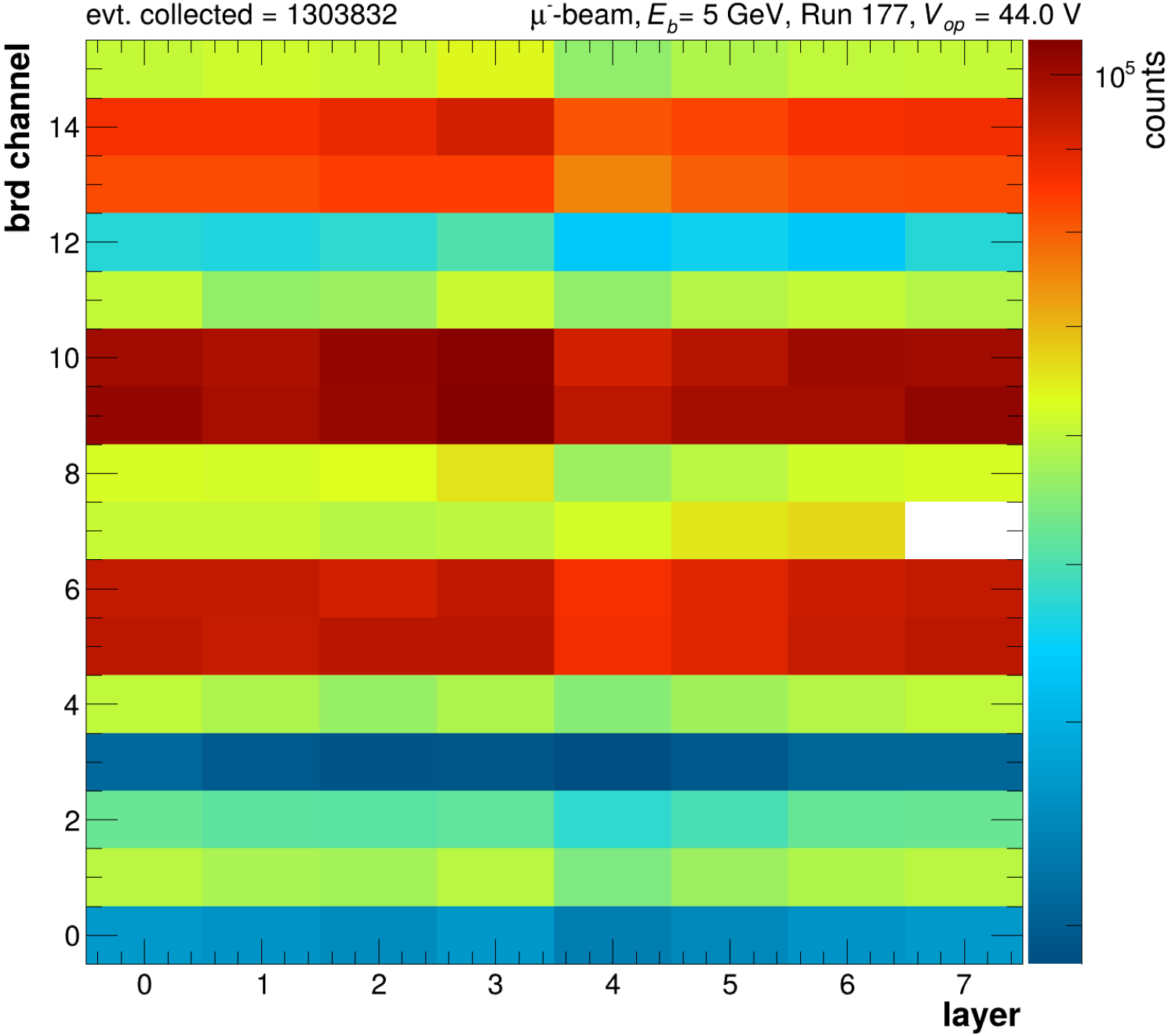

r:1 c:0 m:1
Total events: 958844
Landau-Gauss fit, trigg.
L MPV = 82.23 ± 0.27
Max = 89.98 ± 0.29
χ²/ndf = 20.94
SNR = 2.51
r:1 c:1 m:1
Total events: 952873
Landau-Gauss fit, trigg.
L MPV = 87.63 ± 0.24
Max = 96.42 ± 0.27
χ²/ndf = 43.92
SNR = 2.40
r:1 c:2 m:1
Total events: 948787
Landau-Gauss fit, trigg.
L MPV = 81.18 ± 0.12
Max = 86.04 ± 0.13
χ²/ndf = 11.08
SNR = 2.38
r:1 c:3 m:1 layer 4
Total events: 927048
Landau-Gauss fit, trigg.
L MPV = 95.55 ± 0.19
Max = 102.35 ± 0.20
χ²/ndf = 16.02
SNR = 2.57
μ⁻-beam, E_b = 5 GeV
Run 177, V_op = 44.0 V
r:0 c:0 m:1
Total events: 931370
Landau-Gauss fit, trigg.
L MPV = 91.62 ± 0.28
Max = 100.30 ± 0.31
χ²/ndf = 33.15
SNR = 2.55
r:0 c:1 m:1
Total events: 953365
Landau-Gauss fit, trigg.
L MPV = 81.02 ± 0.19
Max = 89.17 ± 0.21
χ²/ndf = 46.87
SNR = 2.49
r:0 c:2 m:1
Total events: 921454
Landau-Gauss fit, trigg.
L MPV = 88.95 ± 0.26
Max = 98.93 ± 0.29
χ²/ndf = 52.75
SNR = 2.23
r:0 c:3 m:1
Total events: 907438
Landau-Gauss fit, trigg.
L MPV = 95.44 ± 0.19
Max = 101.90 ± 0.20
χ²/ndf = 11.93
SNR = 2.25
r:1 c:0 m:0
Total events: 1299094
Landau-Gauss fit, trigg.
L MPV = 84.99 ± 0.25
Max = 93.17 ± 0.27
χ²/ndf = 30.67
SNR = 2.37
r:1 c:1 m:0
Total events: 1295186
Landau-Gauss fit, trigg.
L MPV = 84.64 ± 0.22
Max = 93.35 ± 0.24
χ²/ndf = 50.13
SNR = 2.44
r:1 c:2 m:0
Total events: 1294872
Landau-Gauss fit, trigg.
L MPV = 84.53 ± 0.12
Max = 89.91 ± 0.13
χ²/ndf = 15.30
SNR = 2.34
r:1 c:3 m:0
Total events: 1294812
Landau-Gauss fit, trigg.
L MPV = 87.61 ± 0.27
Max = 97.06 ± 0.30
χ²/ndf = 38.13
SNR = 2.49
r:0 c:0 m:0
Total events: 1298334
Landau-Gauss fit, trigg.
L MPV = 85.93 ± 0.24
Max = 92.89 ± 0.26
χ²/ndf = 11.38
SNR = 2.24
r:0 c:1 m:0
Total events: 1298142
Landau-Gauss fit, trigg.
L MPV = 95.06 ± 0.21
Max = 102.66 ± 0.22
χ²/ndf = 18.19
SNR = 2.38
r:0 c:2 m:0
Total events: 1295445
Landau-Gauss fit, trigg.
L MPV = 86.83 ± 0.32
Max = 96.00 ± 0.35
χ²/ndf = 26.95
SNR = 2.90
r:0 c:3 m:0
Total events: 1294876
Landau-Gauss fit, trigg.
L MPV = 89.75 ± 0.24
Max = 95.47 ± 0.25
χ²/ndf = 7.09
SNR = 2.50
counts
ADC (arb. units)

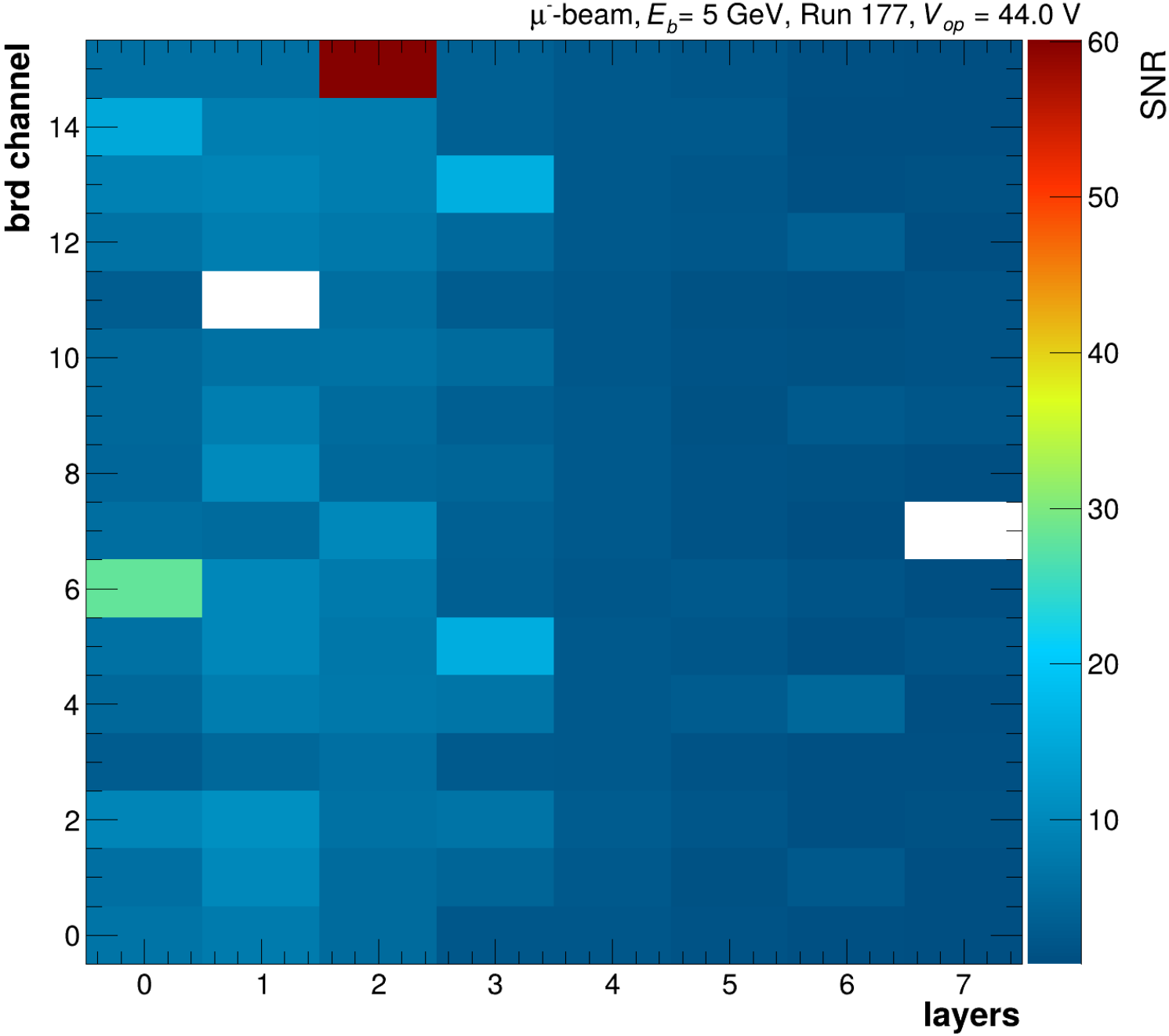

r:1 c:0 m:1
Total events: 943117
Landau-Gauss fit, trigg.
L MPV = 91.32 ± 0.25
Max = 98.51 ± 0.27
χ²/ndf = 16.52
SNR = 2.12
r:1 c:1 m:1
Total events: 943612
Landau-Gauss fit, trigg.
L MPV = 89.49 ± 0.15
Max = 95.03 ± 0.16
χ²/ndf = 15.30
SNR = 1.96
r:1 c:2 m:1
Total events: 840110
Landau-Gauss fit, trigg.
L MPV = 106.09 ± 0.19
Max = 113.85 ± 0.21
χ²/ndf = 22.16
SNR = 2.53
r:1 c:3 m:1 layer 5
Total events: 929014
Landau-Gauss fit, trigg.
L MPV = 94.41 ± 0.34
Max = 103.25 ± 0.37
χ²/ndf = 29.84
SNR = 1.88
μ⁻-beam, E_b = 5 GeV
Run 177, V_op = 44.0 V
r:0 c:0 m:1
Total events: 960224
Landau-Gauss fit, trigg.
L MPV = 97.70 ± 0.25
Max = 105.60 ± 0.27
χ²/ndf = 18.56
SNR = 1.54
r:0 c:1 m:1
Total events: 937190
Landau-Gauss fit, trigg.
L MPV = 91.40 ± 0.14
Max = 97.51 ± 0.15
χ²/ndf = 24.50
SNR = 1.32
r:0 c:2 m:1
Total events: 923014
Landau-Gauss fit, trigg.
L MPV = 99.06 ± 0.31
Max = 108.40 ± 0.34
χ²/ndf = 52.42
SNR = 1.59
r:0 c:3 m:1
Total events: 950983
Landau-Gauss fit, trigg.
L MPV = 98.67 ± 0.23
Max = 105.79 ± 0.24
χ²/ndf = 11.29
SNR = 1.26
r:1 c:0 m:0
Total events: 1298633
Landau-Gauss fit, trigg.
L MPV = 99.76 ± 0.19
Max = 106.32 ± 0.21
χ²/ndf = 15.73
SNR = 2.91
r:1 c:1 m:0
Total events: 1296405
Landau-Gauss fit, trigg.
L MPV = 104.78 ± 0.23
Max = 113.57 ± 0.25
χ²/ndf = 38.70
SNR = 1.97
r:1 c:2 m:0
Total events: 1269936
Landau-Gauss fit, trigg.
L MPV = 89.20 ± 0.14
Max = 94.86 ± 0.15
χ²/ndf = 18.51
SNR = 2.36
r:1 c:3 m:0
Total events: 1297164
Landau-Gauss fit, trigg.
L MPV = 94.17 ± 0.33
Max = 103.62 ± 0.37
χ²/ndf = 32.94
SNR = 1.62
r:0 c:0 m:0
Total events: 1301920
Landau-Gauss fit, trigg.
L MPV = 100.92 ± 0.41
Max = 109.41 ± 0.45
χ²/ndf = 20.06
SNR = 1.57
r:0 c:1 m:0
Total events: 1298490
Landau-Gauss fit, trigg.
L MPV = 100.57 ± 0.21
Max = 107.42 ± 0.22
χ²/ndf = 17.84
SNR = 1.32
r:0 c:2 m:0
Total events: 1297449
Landau-Gauss fit, trigg.
L MPV = 97.64 ± 0.36
Max = 106.36 ± 0.39
χ²/ndf = 25.53
SNR = 1.94
r:0 c:3 m:0
Total events: 1301369
Landau-Gauss fit, trigg.
L MPV = 100.28 ± 0.28
Max = 106.81 ± 0.30
χ²/ndf = 6.46
SNR = 1.50
counts
ADC (arb. units)

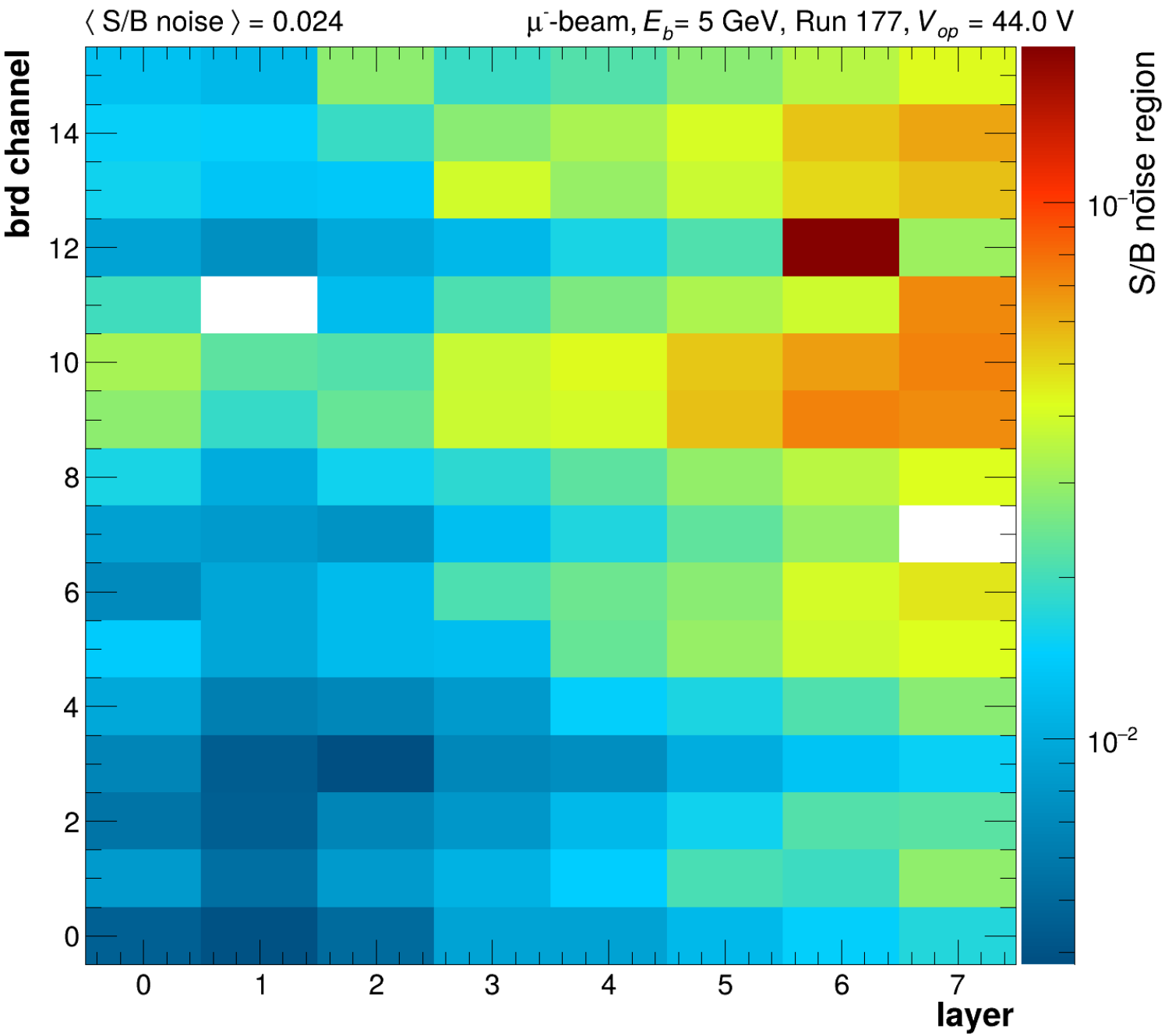

r:1 c:0 m:1
Total events: 179478
Landau-Gauss fit, trigg.
L MPV = 118.79 ± 0.36
Max = 127.89 ± 0.39
χ²/ndf = 15.06
SNR = 3.41
r:1 c:1 m:1
Total events: 948190
Landau-Gauss fit, trigg.
L MPV = 114.02 ± 0.25
Max = 122.58 ± 0.27
χ²/ndf = 16.54
SNR = 0.98
r:1 c:2 m:1
Total events: 934277
Landau-Gauss fit, trigg.
L MPV = 106.12 ± 0.22
Max = 113.95 ± 0.23
χ²/ndf = 21.93
SNR = 0.95
r:1 c:3 m:1 layer 6
Total events: 949366
Landau-Gauss fit, trigg.
L MPV = 118.82 ± 0.31
Max = 127.85 ± 0.33
χ²/ndf = 12.97
SNR = 1.15
μ-beam, E_b = 5 GeV
Run 177, V_{op} = 44.0 V
r:0 c:0 m:1
Total events: 959801
Landau-Gauss fit, trigg.
L MPV = 111.63 ± 0.31
Max = 114.86 ± 0.32
χ²/ndf = 27.98
SNR = 1.25
r:0 c:1 m:1
r:0 c:2 m:1
Total events: 942762
Landau-Gauss fit, trigg.
L MPV = 110.83 ± 0.22
Max = 119.03 ± 0.24
χ²/ndf = 20.19
SNR = 1.19
r:0 c:3 m:1
Total events: 949380
Landau-Gauss fit, trigg.
L MPV = 111.09 ± 0.30
Max = 119.42 ± 0.32
χ²/ndf = 10.50
SNR = 0.95
r:1 c:0 m:0
Total events: 1023825
Landau-Gauss fit, trigg.
L MPV = 105.66 ± 0.27
Max = 113.63 ± 0.29
χ²/ndf = 19.95
SNR = 4.71
r:1 c:1 m:0
Total events: 1299572
Landau-Gauss fit, trigg.
L MPV = 113.42 ± 0.22
Max = 121.97 ± 0.23
χ²/ndf = 24.39
SNR = 1.10
r:1 c:2 m:0
Total events: 1297714
Landau-Gauss fit, trigg.
L MPV = 111.93 ± 0.30
Max = 121.33 ± 0.33
χ²/ndf = 41.13
SNR = 1.80
r:1 c:3 m:0
Total events: 1301089
Landau-Gauss fit, trigg.
L MPV = 104.34 ± 0.25
Max = 112.11 ± 0.27
χ²/ndf = 9.65
SNR = 0.94
r:0 c:0 m:0
Total events: 1302574
Landau-Gauss fit, trigg.
L MPV = 107.20 ± 0.48
Max = 115.25 ± 0.52
χ²/ndf = 14.29
SNR = 1.08
r:0 c:1 m:0
Total events: 1208062
Landau-Gauss fit, trigg.
L MPV = 104.11 ± 0.27
Max = 111.92 ± 0.29
χ²/ndf = 18.10
SNR = 2.54
r:0 c:2 m:0
Total events: 1299753
Landau-Gauss fit, trigg.
L MPV = 108.48 ± 0.28
Max = 116.43 ± 0.31
χ²/ndf = 14.30
SNR = 1.15
r:0 c:3 m:0
Total events: 1301797
Landau-Gauss fit, trigg.
L MPV = 107.80 ± 0.38
Max = 115.85 ± 0.41
χ²/ndf = 7.37
SNR = 0.96
counts
ADC (arb. units)

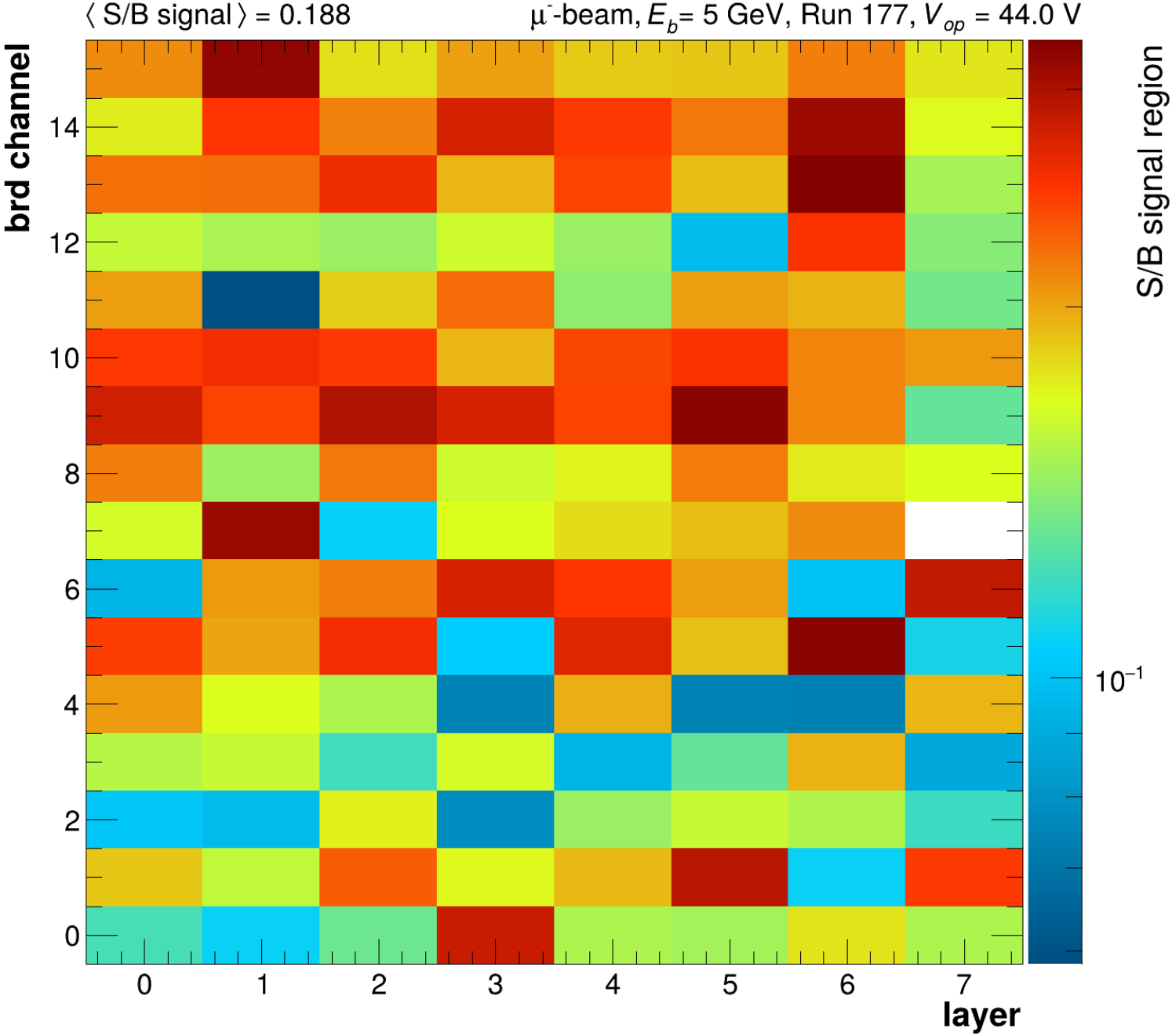

r:1 c:0 m:1
Total events: 960624
Landau-Gauss fit, trigg.
L MPV = 117.48 ± 0.69
Max = 116.98 ± 0.68
χ²/ndf = 19.22
SNR = 0.88
r:1 c:1 m:1
Total events: 943012
Landau-Gauss fit, trigg.
L MPV = 120.15 ± 0.31
Max = 129.27 ± 0.33
χ²/ndf = 18.40
SNR = 1.20
r:1 c:2 m:1
Total events: 934864
Landau-Gauss fit, trigg.
L MPV = 117.02 ± 0.29
Max = 125.94 ± 0.31
χ²/ndf = 18.37
SNR = 0.89
r:1 c:3 m:1 layer 7
Total events: 923497
Landau-Gauss fit, trigg.
L MPV = 123.78 ± 0.42
Max = 133.75 ± 0.45
χ²/ndf = 14.58
SNR = 0.87
μ-beam, E_b = 5 GeV
Run 177, V_{op} = 44.0 V
r:0 c:0 m:1
Total events: 951722
Landau-Gauss fit, trigg.
L MPV = 116.37 ± 0.33
Max = 116.01 ± 0.33
χ²/ndf = 22.51
SNR = 0.87
r:0 c:1 m:1
Total events: 941606
Landau-Gauss fit, trigg.
L MPV = 125.52 ± 0.31
Max = 135.11 ± 0.33
χ²/ndf = 25.88
SNR = 1.90
r:0 c:2 m:1
Total events: 840430
Landau-Gauss fit, trigg.
L MPV = 129.33 ± 0.36
Max = 139.71 ± 0.39
χ²/ndf = 25.00
SNR = 1.46
r:0 c:3 m:1
Total events: 584693
Landau-Gauss fit, trigg.
L MPV = 133.05 ± 0.48
Max = 143.84 ± 0.52
χ²/ndf = 17.35
SNR = 1.52
r:1 c:0 m:0
Total events: 1302172
Landau-Gauss fit, trigg.
L MPV = 113.11 ± 0.25
Max = 112.99 ± 0.25
χ²/ndf = 26.04
SNR = 0.86
r:1 c:1 m:0
Total events: 1298742
Landau-Gauss fit, trigg.
L MPV = 122.76 ± 0.31
Max = 132.55 ± 0.33
χ²/ndf = 25.24
SNR = 1.68
r:1 c:2 m:0
Total events: 1298382
Landau-Gauss fit, trigg.
L MPV = 124.75 ± 0.32
Max = 134.55 ± 0.34
χ²/ndf = 27.68
SNR = 0.74
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 1302193
Landau-Gauss fit, trigg.
L MPV = 113.76 ± 0.45
Max = 122.56 ± 0.48
χ²/ndf = 7.80
SNR = 0.81
r:0 c:1 m:0
Total events: 1299833
Landau-Gauss fit, trigg.
L MPV = 119.68 ± 0.39
Max = 129.09 ± 0.42
χ²/ndf = 13.79
SNR = 0.72
r:0 c:2 m:0
Total events: 1272138
Landau-Gauss fit, trigg.
L MPV = 136.20 ± 0.09
Max = 137.77 ± 0.09
χ²/ndf = 21.41
SNR = 1.38
r:0 c:3 m:0
Total events: 1167593
Landau-Gauss fit, trigg.
L MPV = 118.12 ± 0.54
Max = 127.49 ± 0.58
χ²/ndf = 8.56
SNR = 1.09
counts
ADC (arb. units)

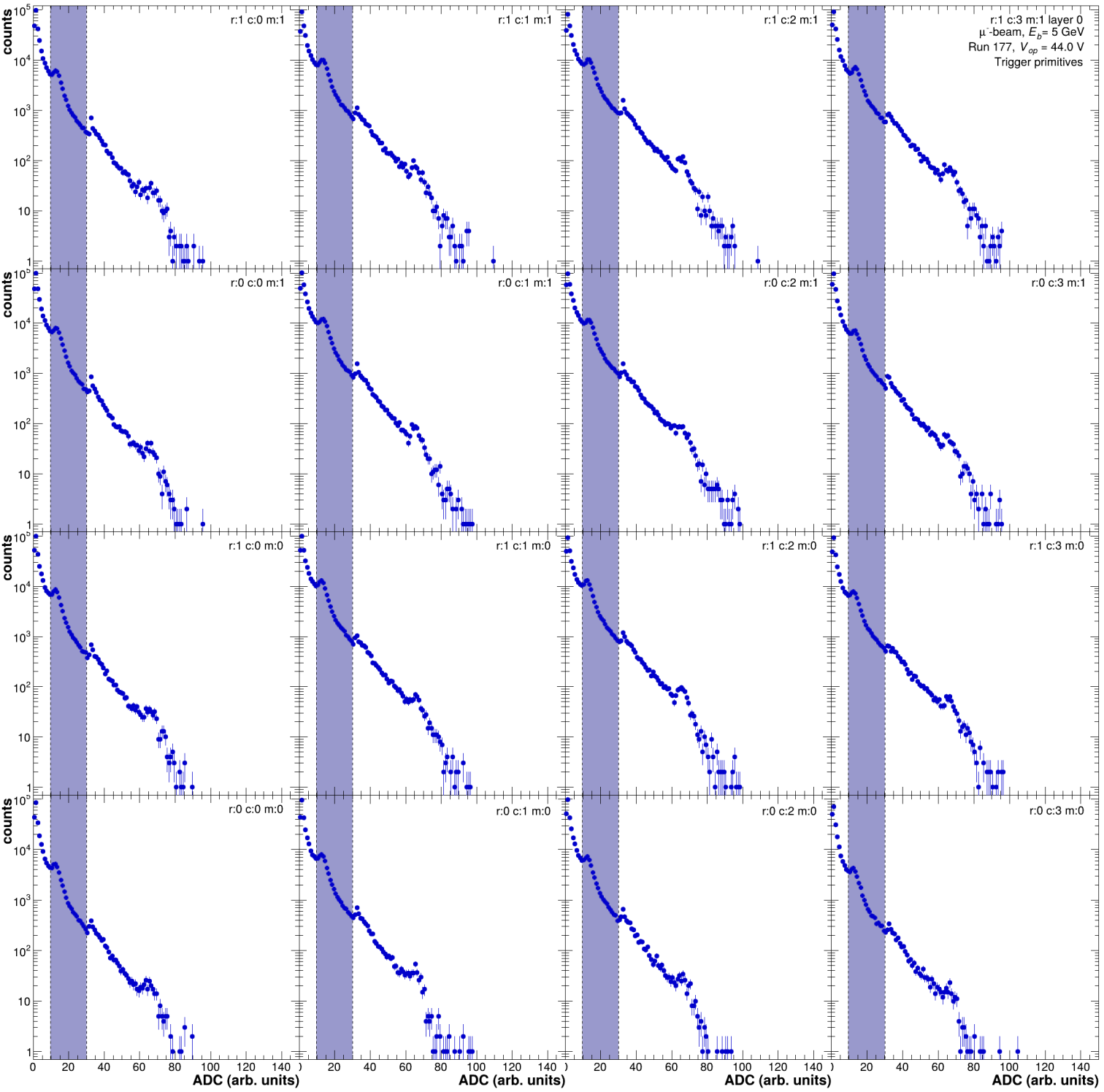

μ⁻-beam, E_b= 5 GeV, Run 177, V_{op} = 44.0 V
Y (row)
X (col)
Num triggers
×10³
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5
800
700
600
500
400
300

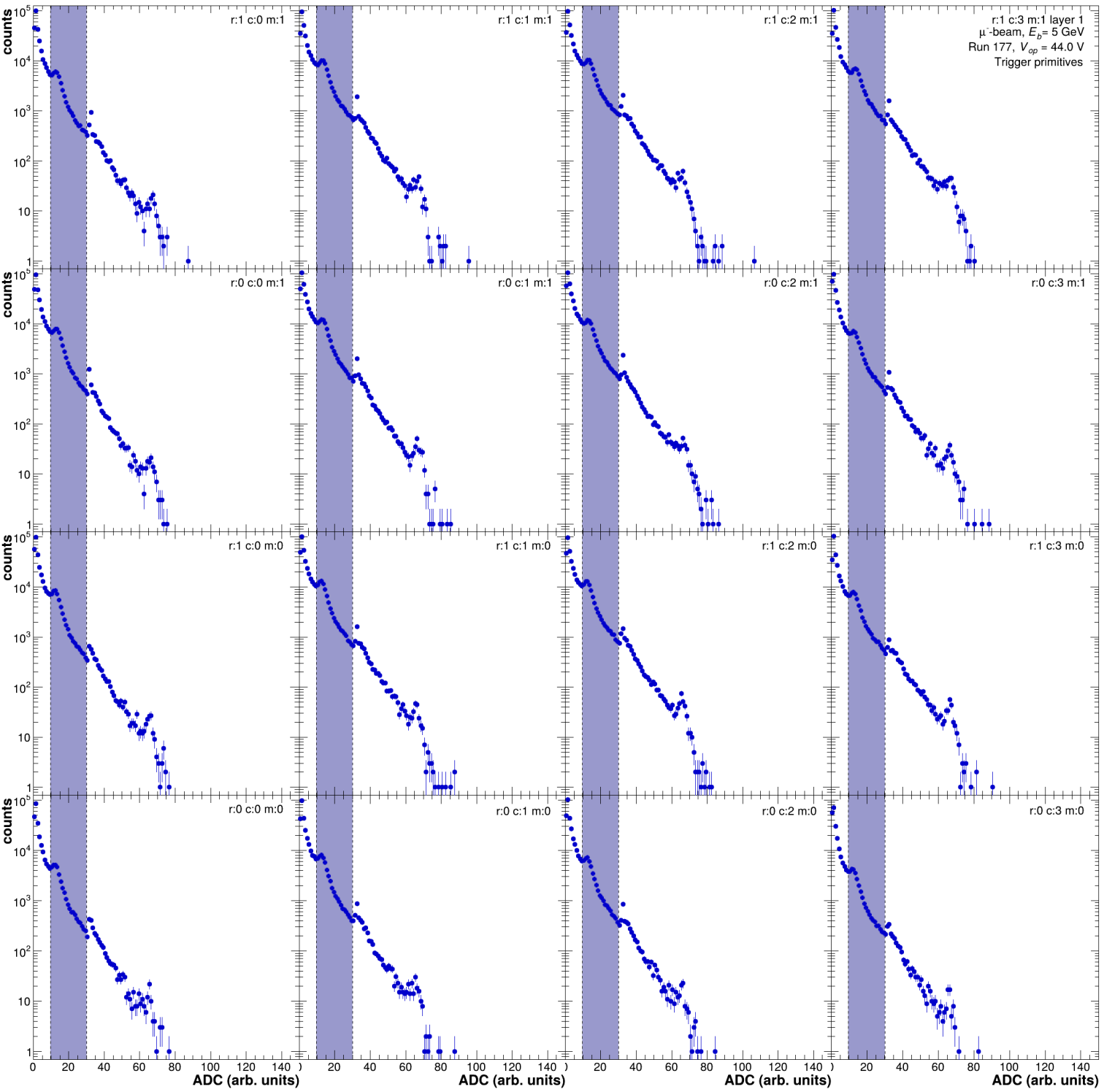

evt. collected = 1303832
μ⁻-beam, E_b= 5 GeV, Run 177, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
10^5
counts

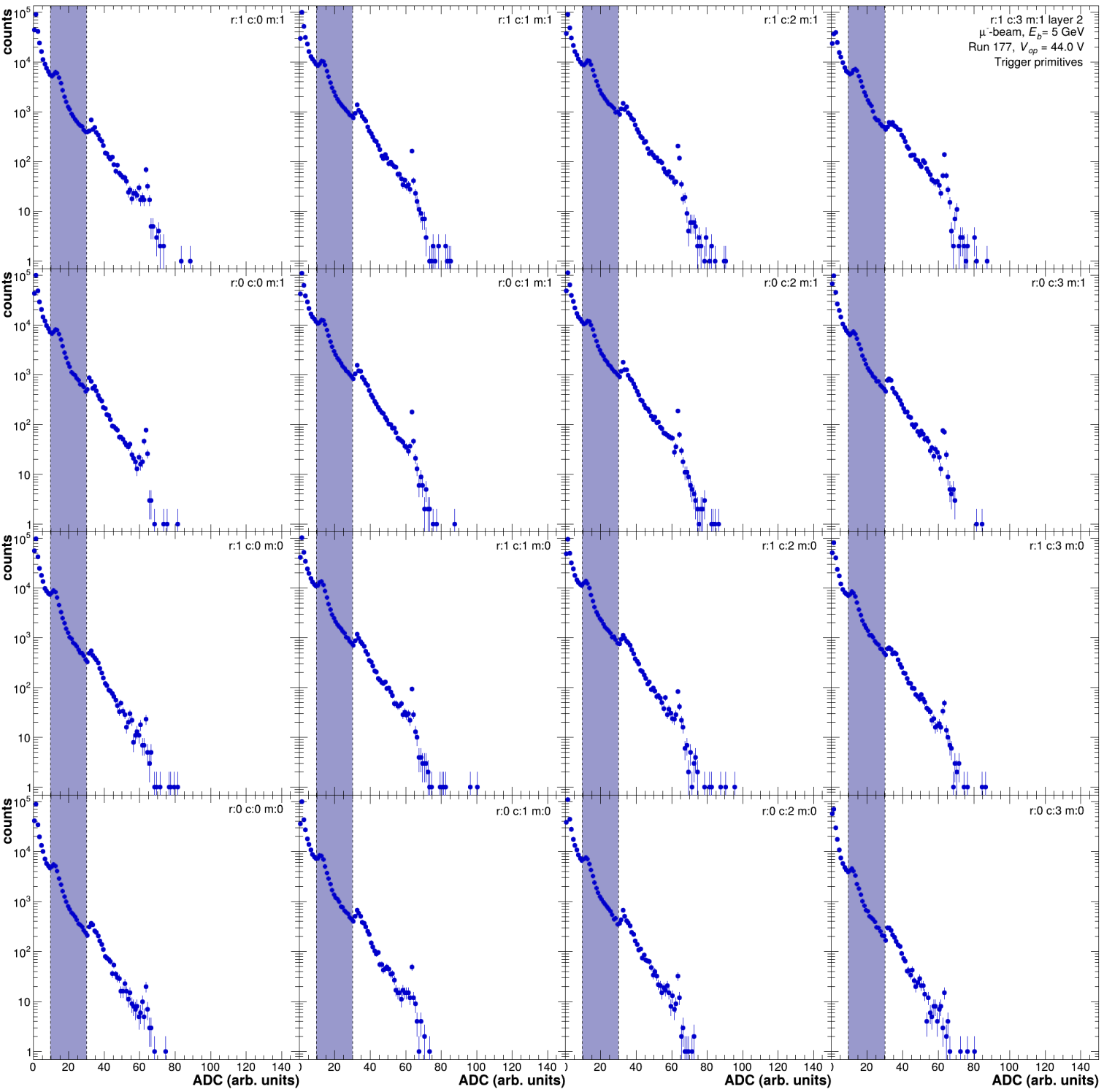

μ⁻-beam, E_b= 5 GeV, Run 177, V_{op} = 44.0 V
brd channel
layers
SNR
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14
10
20
30
40
50
60

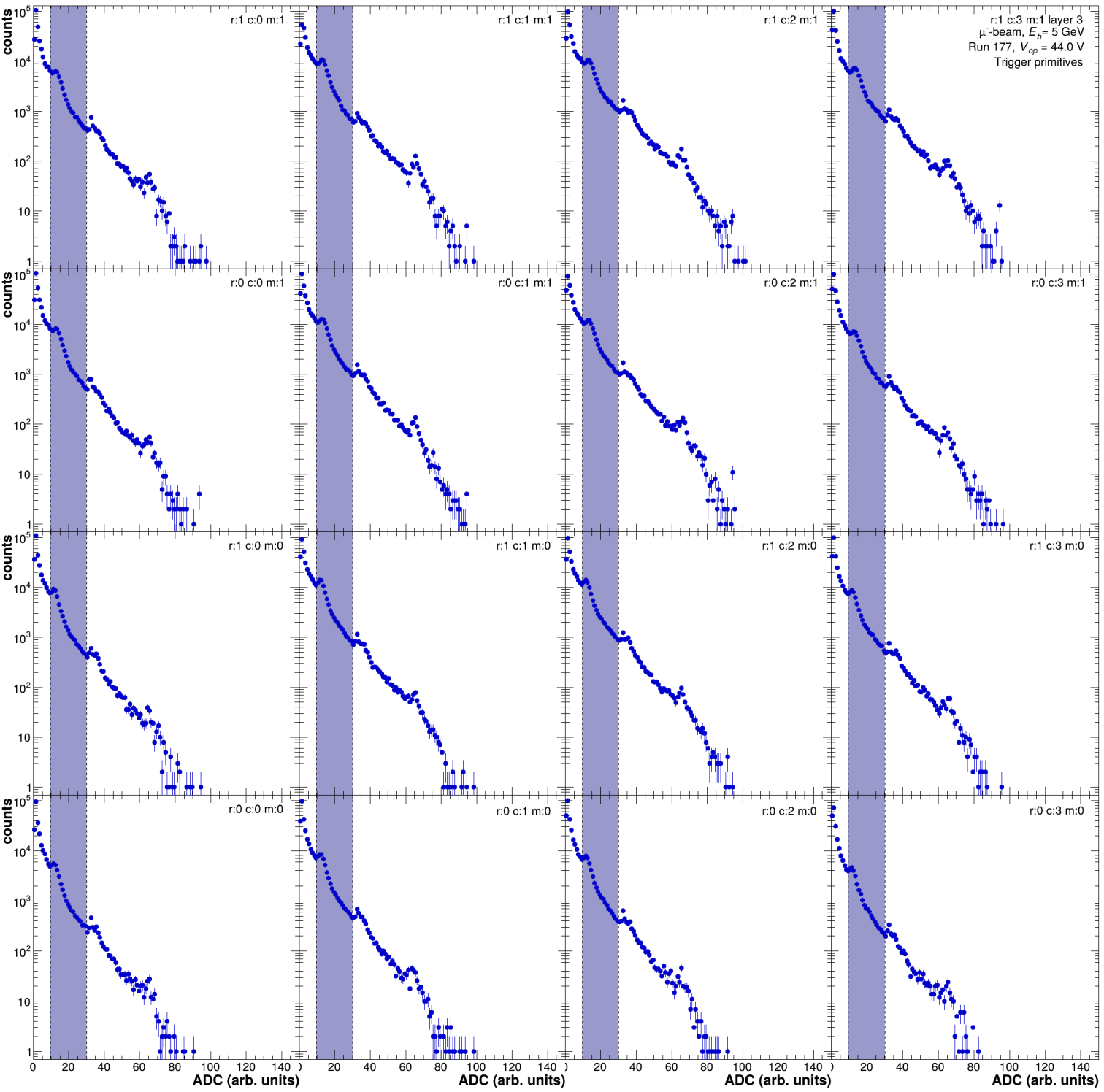

⟨ S/B noise ⟩ = 0.024
μ⁻-beam, E_b= 5 GeV, Run 177, V_{op} = 44.0 V
brd channel
layer
S/B noise region
10^{-1}
10^{-2}
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7

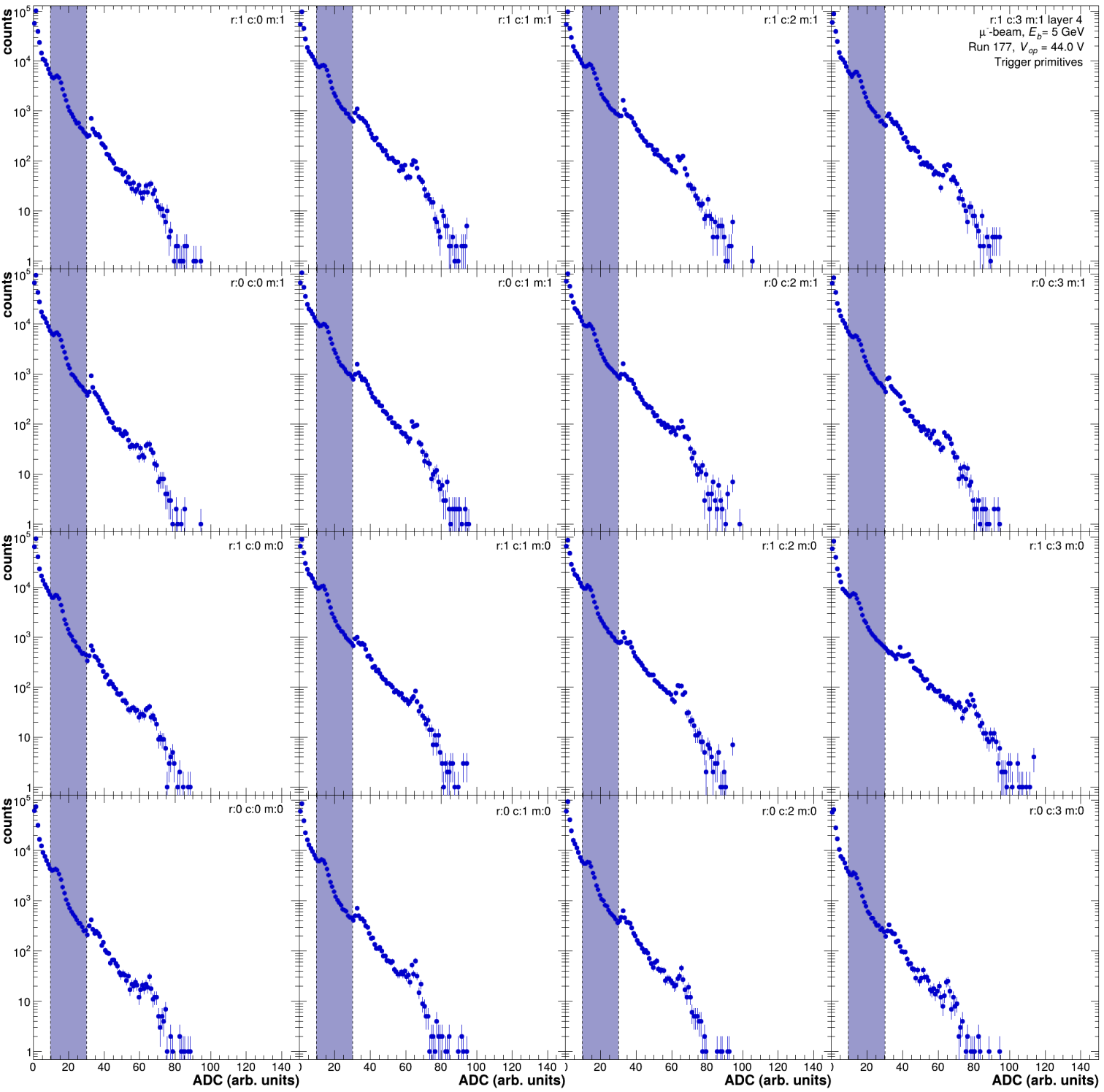

⟨ S/B signal ⟩ = 0.188
μ⁻-beam, E_b= 5 GeV, Run 177, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B signal region
10^{-1}

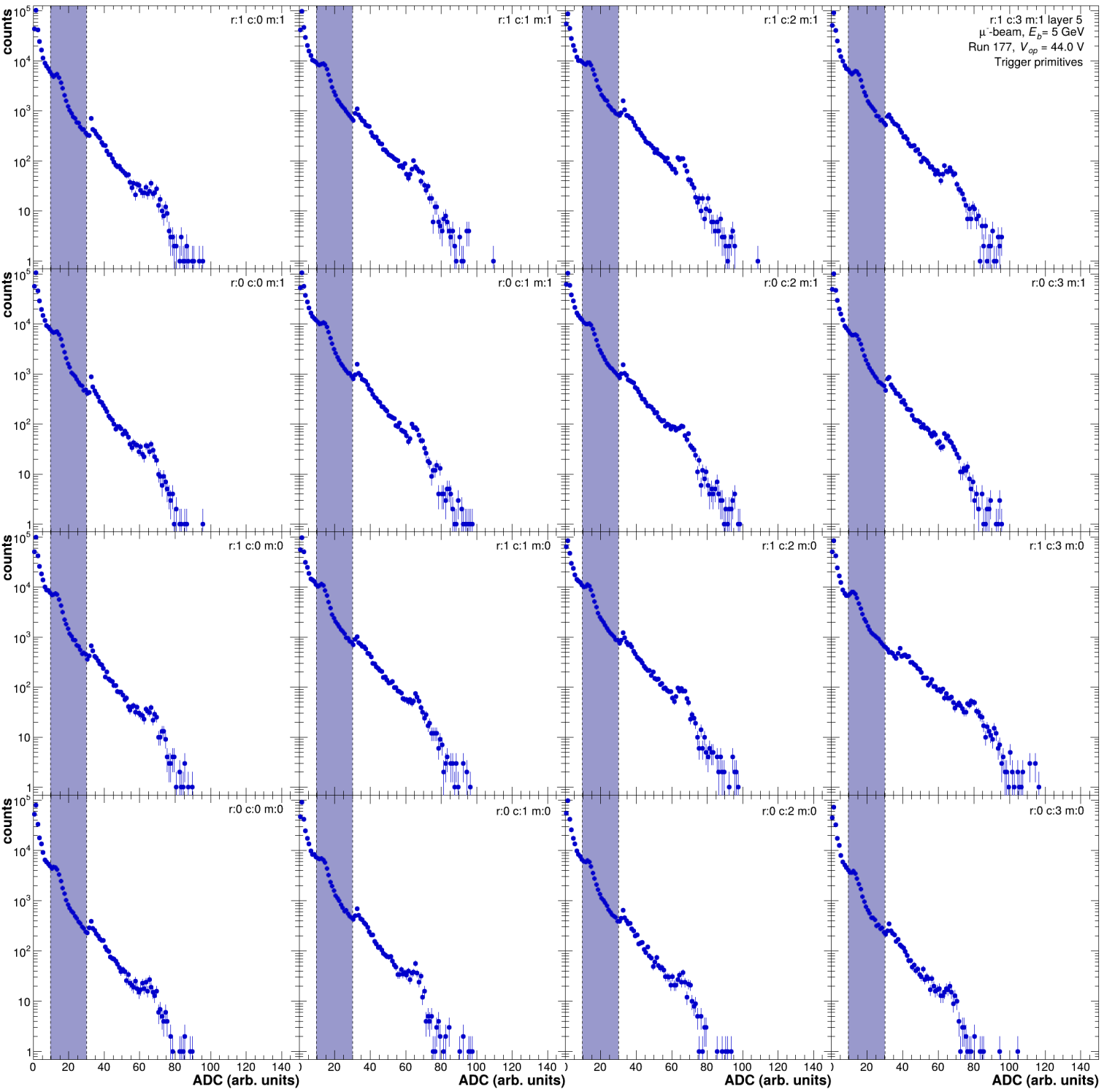

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 177, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

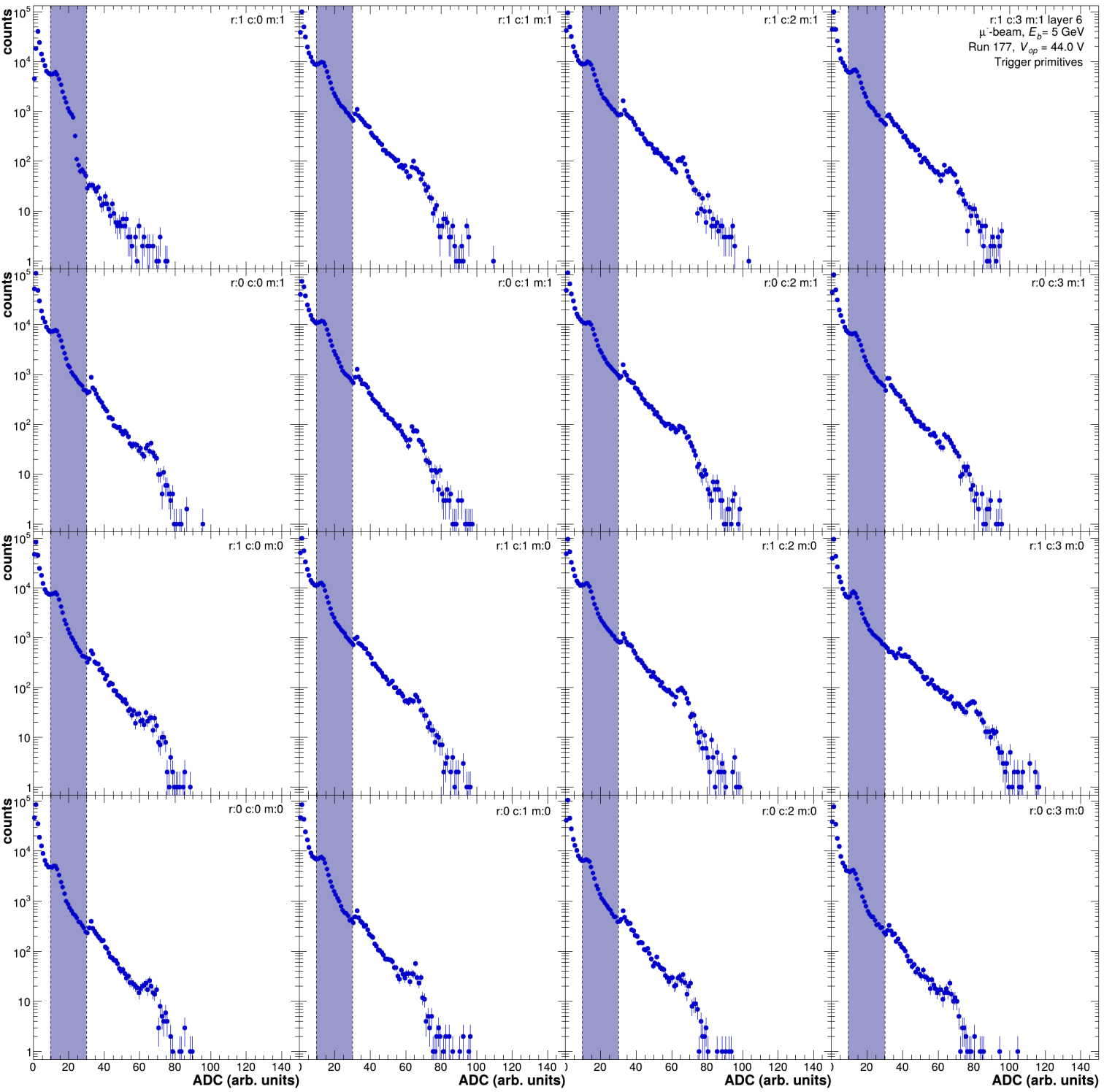

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 177, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

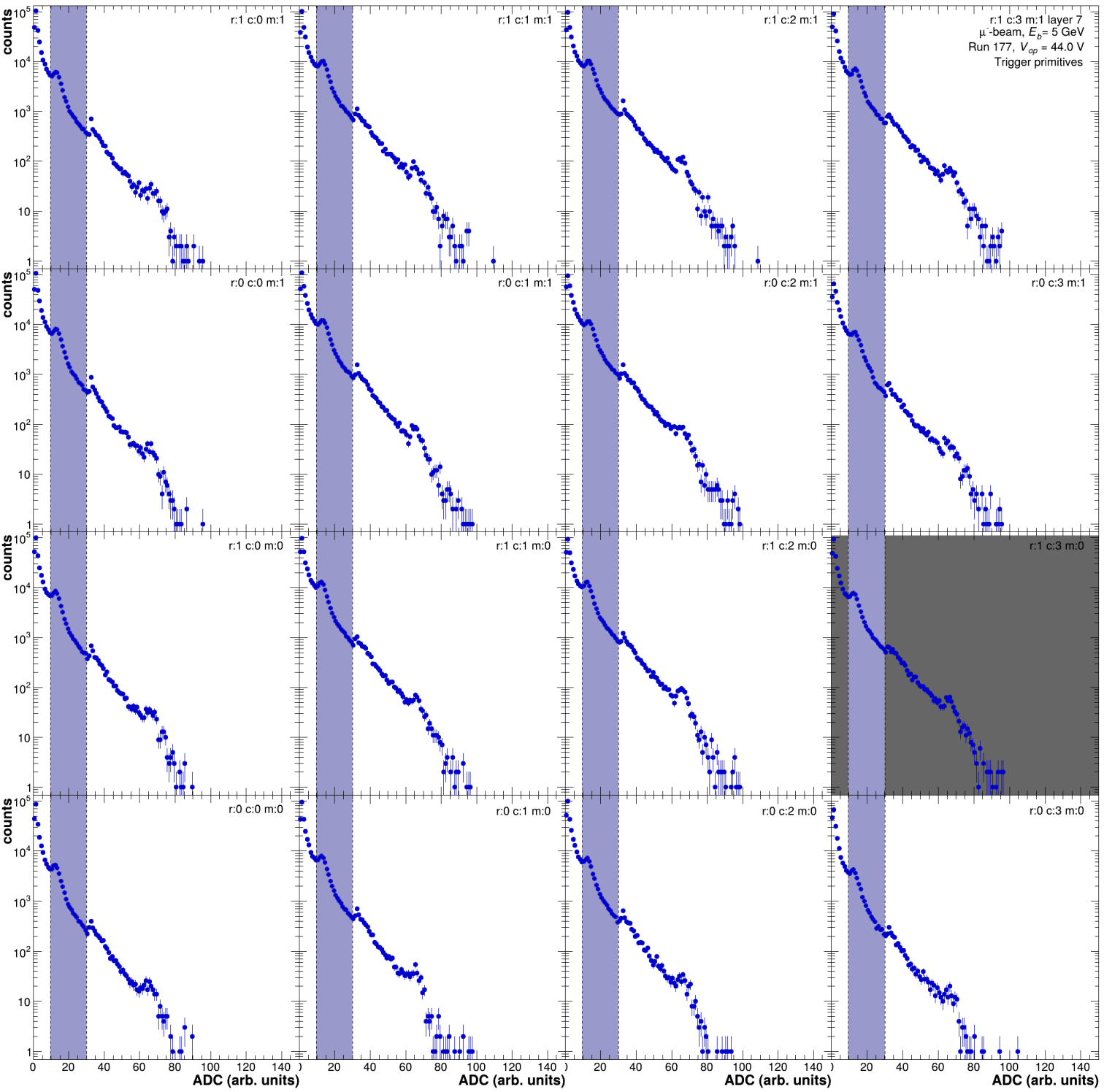

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 177, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 177, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 177, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 177, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 177, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 177, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)